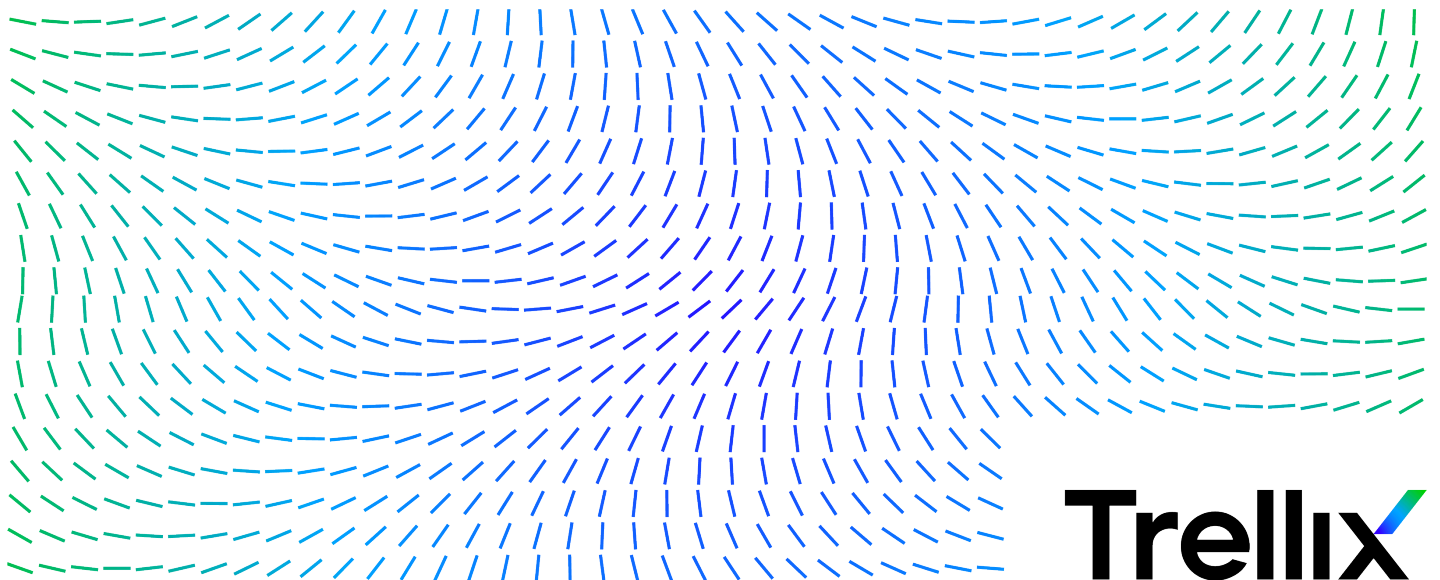


Trellix ePolicy Orchestrator - On-prem 5.10.0 Installation Guide

Last Updated: February 23, 2026



Contents

Installation overview	8
Which type of installation do you need?	8
Single server installation workflow	8
Cloud services installation workflow	9
Cluster installation workflow	10
Upgrade installation workflow	11
Planning your installation	13
Considerations for scalability	13
Baseline calculations	17
Sizing distributed repositories	20
Sizing DXL Brokers	21
Combining servers	22
Factors that affect ePO - On-prem performance	23
Internet protocols in a managed environment	24
Things to know before installation	25
System requirements	26
System requirements and recommendations	26
Software requirements and recommendations	27
Operating system requirements	29
Supported virtual infrastructure software	31
Supported SQL Servers	31
Disable Backup Retention Period in Amazon Relational Database Service for SQL Server	32
Configure TCP/IP access to the SQL Server	33
Supported Internet browsers	34

Agent Handler server requirements	35
SQL Server installation documented in this guide	36
Required SQL permissions	36
Supported SQL database user name and password formats	37
Port options	38
Automatic product installation	39
Distributed repository requirements	40
Supported products and known issues	40
Installing ePO - On-prem 5.10.0 on a single server	42
Download ePO - On-prem 5.10.0 installation packages	42
Things to know before installation	43
Install ePO - On-prem on a single server	43
Set recovery model in SQL server	48
Installing Trellix ePO on a cloud server	50
Using a Microsoft Azure server for ePO - On-prem	50
Port requirements	51
Configure the Microsoft Azure server for ePO - On-prem	52
Install ePO - On-prem on an Azure server	53
Update ePO - On-prem public DNS name	53
Manage your Agent Handlers	53
Distributed Repository connections	54
Using a Google Cloud Platform (GCP) server for ePO - On-prem	54
Port requirements	56
Configure GCP on ePO - On-prem	56
Install ePO - On-prem on a GCP server	57
Update ePO - On-prem public DNS name	58
Manage your Agent Handlers in the GCP environment	58
Distributed Repository connections	59
Installing ePO on AWS server	59
Create an EC2 instance in Virtual Private Cloud	59

Connect to EC2 instance.	61
Things to do before installation	61
Install ePO on AWS server	61
Manage your Agent Handlers	62
Port requirements.	63
Update ePO - On-prem public DNS name	64
Importing Certificate.	65
Updating host file	65
Installing ePO - On-prem in a cluster environment	66
Create the ePO - On-prem application role.	67
Create the Client Access Point.	68
Add the data drive.	68
Install ePO - On-prem software on each cluster node	68
Create the Generic Service resources	72
Test the ePO - On-prem cluster installation	73
Enabling Multi-Subnet failover support using ePO 5.10.0	73
Setting up your ePO - On-prem 5.10.0 environment	75
Install products automatically on your ePO - On-prem server.	75
Install products manually on your ePO - On-prem server	76
Add systems to the System Tree.	77
Define proxy settings	78
Enable ePO - On-prem license.	78
Confirm that your protection software stops a sample threat	78
Confirm the threat response in ePO - On-prem.	79
Confirm that your systems are being managed by ePO	80
Upgrade to ePO - On-prem 5.10.0 Service Pack 1	81
Before you upgrade	82
Pre-upgrade checklist.	82
Mandatory prerequisites and upgrade requirements.	82
Planning your ePO - On-prem upgrade	83

Download the tools and software you need to upgrade	83
Checking your system for compatibility using the PIA tool	84
Upload Product Compatibility List (PCL) file manually	85
Back up ePO - On-prem databases and directories	85
Update your database server certificates	85
Day of upgrade	86
Upgrade checklist	86
Stop Agent Handler and ePO - On-prem services	86
Run the InstallShield Wizard	87
Upgrade your Agent Handlers	89
Upgrade your ePO - On-prem cluster server	90
After you upgrade	91
Post-upgrade checklist	91
Migrate Certificate Authority Hashing Algorithm from SHA-1 to SHA-2 or higher	91
ePO - On-prem cumulative updates	93
Cumulative update installation procedures	94
Install the latest update on your ePO - On-prem 5.10.0 server	94
Install the latest build on your ePO - On-prem server	96
Repair the latest update on your ePO - On-prem server	97
Install the latest update on Remote Agent Handlers	97
Verify the update installation	98
Roll back updates	98
Use Silent Mode to install a ePO - On-prem update	99

Troubleshooting installation 100

Troubleshooting and log file reference	100
Common installation messages with their causes and solutions	100
Log files for troubleshooting	103
Installer logs	104
Server logs	106
Trellix Agent logs	109

Access your ePO - On-prem with your ePO - SaaS account.....	113
Using Trellix ePO - SaaS Registration token	113
Download and install ePO - SaaS Cloud Bridge	113
Configuring ePO - SaaS Cloud Bridge using Registration token.....	114
Using Trellix ePO - SaaS credentials authentication	116
Download and install ePO - SaaS Cloud Bridge	116
Configure using ePO - SaaS credentials authentication	117
Error messages	119
Adding an SSL certificate to trusted collection	121
Replace the server certificate	121
Install the security certificate for Internet Explorer.....	122
Install the security certificate for Firefox.....	123
Install Agent Handlers	124
Restoring Trellix ePO-On-prem from a Disaster Recovery Snapshot	126
Disaster Recovery Snapshot prerequisites	126
Restore ePO - On-prem software in a single-server environment.....	126
Restore ePO - On-prem software in a cluster environment.....	129
Restore Agent Handler connections.....	131
Using Trellix ePO-On-prem in FIPS mode.....	133
FIPS basics	133
ePO - On-prem operating modes	135
Install ePO - On-prem in FIPS mode.....	136
Upgrade from an earlier FIPS-compliant ePO - On-prem server	136
Restoring ePO - On-prem server in FIPS mode.....	137
Verify FIPS 140-2 operational status in ePO	137
Remove the software.....	139
Uninstall ePO - On-prem	139
Uninstall ePO - On-prem from a cluster	139

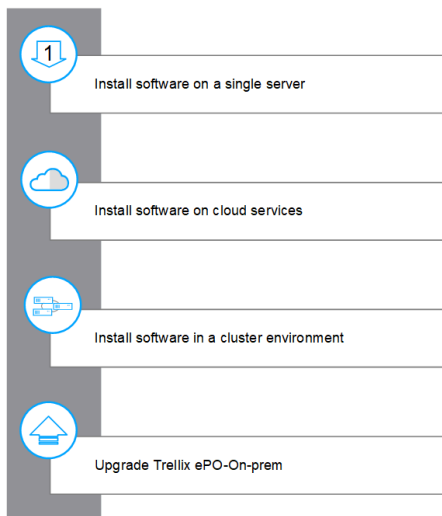
Uninstall Agent Handler from Trellix ePO - On-prem	139
--	-----

Which type of installation do you need?

Install ePO - On-prem software as a single-server installation or as a cluster, cloud, or upgrade installation.

Each installation scenario includes a workflow and procedure. Planning your installation and reviewing system requirements are also part of the installation process.

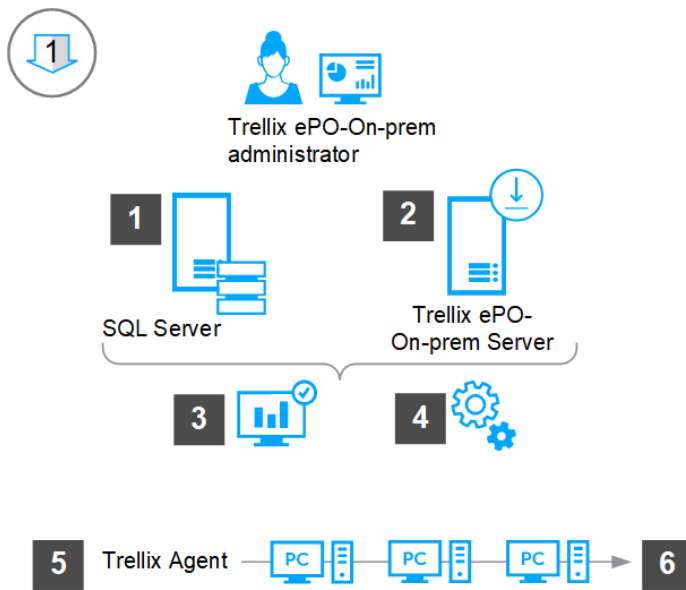
Which type of installation do you need?



Single server installation workflow

Before you can install ePO - On-prem software for the first time, you must ensure your SQL Server software is configured for TCP/IP access and install a supported operating system on the ePO - On-prem server.

1. Ensure your SQL Server is configured for TCP/IP access.
2. Download and extract the ePO - On-prem software from <https://www.trellix.com/en-us/downloads/my-products.html> or the Trellix download site using a grant number.
3. Verify the latest Microsoft updates are running on the SQL Server and the ePO - On-prem server.
4. Run the setup utility on the ePO - On-prem server to install ePO - On-prem. As part of the installation process the ePO - On-prem Pre-Installation Auditor checks for compliance issues.
5. Choose a deployment method to deploy Trellix Agent.
6. Confirm that systems are managed by ensuring that Trellix Agent can successfully connect to ePO - On-prem.

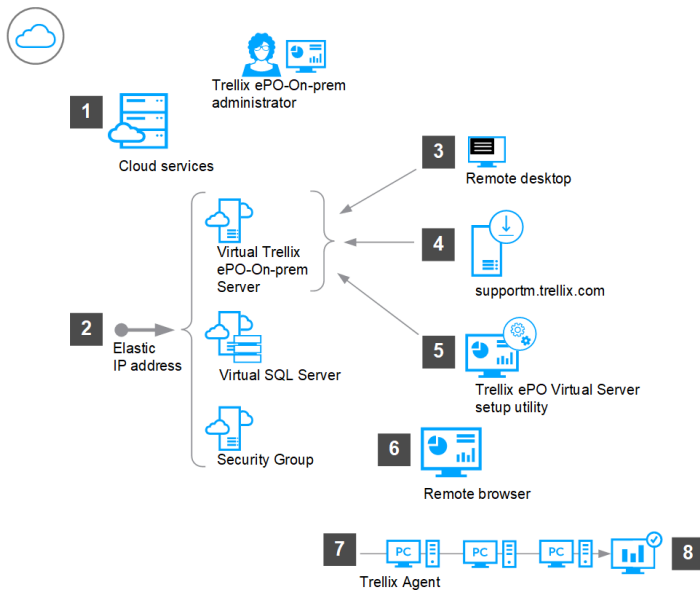


Cloud services installation workflow

Set up a cloud services account and configure your virtual environment to run cloud services with ePO - On-prem.

- Set up a cloud services account and configure these items:
 - Virtual server to use as your ePO - On-prem server
 - Virtual SQL Server
 - Security Group
- Assign an elastic IP address to each virtual server.
- From a management computer, use Remote Desktop to connect to the virtual ePO - On-prem server.
- From Trellix.com, copy the ePO - On-prem software to the virtual ePO - On-prem server.
- From the ePO - On-prem server, run the setup utility.
- Using a remote browser, log on to ePO - On-prem using `https://< elastic IP / DNS of virtual ePO - On-prem server >:<port>`.
 - Update Trellix ePO Server Public DNS in Server Settings with elastic IP address or DNS of virtual ePO - On-prem server.
 - Update Published DNS name or the IP address of Agent Handler (if any) with elastic IP address or DNS of virtual Agent Handler server.

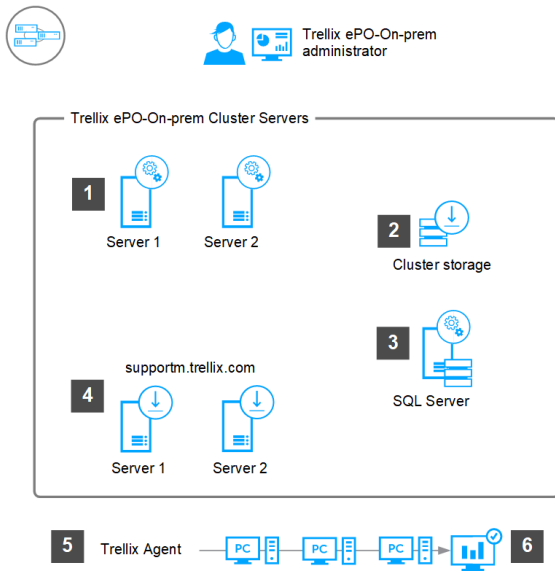
- Create a Trellix Agent deployment URL or extract the Trellix Agent deployment package.
7. Choose a deployment method to deploy Trellix Agent.
 8. Confirm that systems are managed by ensuring that Trellix Agent can successfully connect to ePO - On-prem.



Cluster installation workflow

ePO - On-prem provides high availability for server clusters with Microsoft Cluster Server (MSCS) software.

1. Install Microsoft Cluster Server (MSCS) software on all your servers and configure these items:
 - Shared data drive
 - Quorum drive
 - Failover group
2. Configure shared storage.
3. Configure SQL Server and database settings.
4. Download and install ePO - On-prem software on all servers.
5. Choose a deployment method to deploy Trellix Agent.
6. Confirm that systems are managed by ensuring that Trellix Agent can successfully connect to ePO - On-prem.



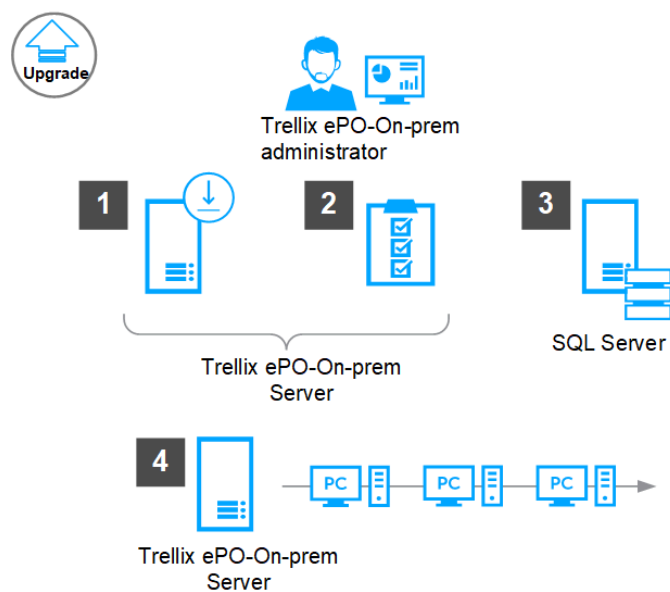
Upgrade installation workflow

Upgrade your existing ePO - On-prem software to a new version.

1. Download and extract the software to your ePO - On-prem server.
2. Prepare the ePO - On-prem server environment.

The ePO - On-prem Pre-Installation Auditor runs, checking compliance with all requirements.

3. Configure SQL Server and database settings.
4. From the ePO - On-prem server, run the setup utility.



Planning your installation

To use your ePO server effectively, create a comprehensive plan specific to your environment.

Considering the unique needs of your environment in advance can reduce the time it takes to get started.

- How many systems do you manage?
- Are your systems located in one network or multiple geographic areas?
- Do you have specific security needs, such as a firewall?
- Do you use Network Address Translation (NAT) in an external network?
- Do you have bandwidth restrictions to remote network segments?
- Do you manage laptops that are connected to the Internet and outside the corporate network?
- Do you have multiple administrators with different permissions across different products, groups of systems, or different functions within the management console?

Considerations for scalability

You can scale your ePO - On-prem infrastructure in multiple ways. Scaling is needed if the environment managed by ePO - On-prem is growing beyond the capacity of the current ePO - On-prem infrastructure.

To grow your ePO - On-prem infrastructure, you can: Move the ePO - On-prem SQL database to a larger and more powerful SQL Server, add more Agent Handlers, or increase CPU and memory to improve storage performance on the SQL Server.

With ePO - On-prem software, you can scale your network vertically or horizontally.

- **Vertical scalability** — Adding and upgrading to bigger, faster hardware to manage larger and larger environments. Scaling vertically is accomplished by upgrading your server hardware, and installing ePO - On-prem on multiple servers throughout your network, each with its own database.
- **Horizontal scalability** — Increasing the size of the environment that one ePO - On-prem server can manage. Scaling horizontally is accomplished by installing additional Agent Handlers, all sharing a single database.

Make sure the ePO - On-prem infrastructure is scaled to handle major peaks in outbreak situations.

Managed systems and servers

The number of systems your ePO - On-prem server manages dictates the number and size of the servers needed. It also dictates the recommended server sizing needed to manage these systems.

Option	< 1,500 systems	1,500–10,000 systems	10,000–25,000 systems	25,000–75,000 systems	> 75,000 systems
Virtual ePO - On-prem server	Yes	Yes	Yes	Yes	Yes
Virtual SQL database server	Yes	Yes	Yes	Conditional*	Conditional*
ePO - On-prem server and SQL database on the same server	Conditional*	Conditional*	Conditional*	Conditional*	Not recommended
Add distributed repositories	Not required	Conditional*	Conditional*	Yes	Yes
Add Agent Handlers (virtual)	Not required	Conditional*	Conditional*	Yes	Yes

*For more information, see [Combining servers](#).



Tip

We recommend one Agent Handler for every 50,000 systems.

There is no hard limit on the number of systems ePO - On-prem can manage. The primary limitation is the SQL database performance, specifically disk performance (IOPS – I/Os per second). You can scale the SQL database, add distributed repositories, and add Agent Handlers to manage more systems as needed.

Important sizing considerations

- **Environment** — Estimates based on a ePO - On-prem server running the Trellix ENS products.
- **Operating systems** — You must use a 64-bit operating system for the ePO - On-prem server and SQL Server.
- **CPU cores** — Server class, minimum 2.2 GHz.
- **RAM** — Add 16 GB of RAM to SQL Server for every 25,000 nodes.
- **Storage capacity** — Estimated event retention period of 6 months.
- **Storage performance** — Storage estimated event retention period of 6 months.

Recommended hardware based on number of managed systems

Node count	ePO - On-prem server		
	CPU cores	RAM (GB)	Storage (GB)
< 10,000	4	8	300
10,000–25,000	4	8–16	500
25,000–75,000	8	16–32	500
75,000–150,000	12	16–64	500
150,000 +	16	16–64	500

Node count	Agent Handler				
	Number of Agent Handlers	CPU cores	RAM (GB)	Storage (GB)	Notes
< 10,000	—	—	—	—	You can use a single server or multiple VMs.
10,000–25,000	0–1	4	8	150	
25,000–75,000	0–1	4	8	150	
75,000–150,000	1–3	4	8	150	
150,000 +	3+	4	8	150	

Node count	SQL Server			
	CPU cores	RAM (GB)	Storage (TB)	Performance (IOPS)
< 10,000	4	8–16	0.5–1.0	
10,000–25,000	4	8–16	0.5–1.5	
25,000–75,000	8	16–32	1.0–2.0	>10,000
75,000–150,000	16	32–128	2.0–3.0	>30,000

Node count	SQL Server			
	CPU cores	RAM (GB)	Storage (TB)	Performance (IOPS)
150,000 +	32+	128–256	3.0	>90,000

By default, IOPS is calculated with 4-KB sectors. Microsoft recommends that SQL storage solutions are allocated with a 64-KB sector size.

Baseline calculations

Your product configuration requires different resources than the baseline calculations we provide, which are estimates. Modify your baseline to meet the unique needs of your environment.

For calculation purposes, use 10,000 for *SystemsPerAddlCoreEP0Server* and 12,500 for *NumSystemsPerCoreAgentHandler*. We provide a table of calculations as a starting point, but be aware that your calculations depend on your environment. For example, for an environment with only 10,000 systems, the hardware requirements for a SQL Server using these calculations are high. For smaller or less complex environments, use the Combining Servers page for more accurate estimates.

Component	Calculation	100,000 systems with 6-month retention period and average 10 events/day per system
Estimating cores for ePO - On-prem server	$\text{NumEpoCores} = 4 + (\text{NumSystems} / \text{SystemsPerAddlCoreEP0Server})$	$4 + (100,000 / 10,000) = 14$ ePO - On-prem processor cores
Estimating RAM for ePO - On-prem server	$\text{NumEpoCores} * 4$	$14 * 4 = 56$ GB RAM
Estimating cores for Agent Handlers (every 4 = new	$\text{NumAgentHandlerCores} =$	$(100,000 - 50,000) / 12,500 = 4$ Agent Handler cores (1 Agent Handler)

Component	Calculation	100,000 systems with 6-month retention period and average 10 events/day per system
Agent Handler)	$(\text{NumSystems} - 50,000) / \text{NumSystemsPerCoreAgentHandler}$	
Estimating SQL cores	$(\text{NumEpoCores} + \text{NumAgentHandlerCores}) * 2$	$(14 + 4) * 2 = 36$ SQL cores
Estimating SQL RAM	$8 \text{ GB} * \text{NumSQLCores}$	$8 * 36 = 288$ GB RAM
Estimating SQL storage	$(\text{NumSystems} * \text{AveSizeOfAgentProperties}) + (\text{NumSystems} * \text{EventsPerDay} * \text{AveSizeOfEvent} * \text{NumDaysToRetain})$	$(100,000 * 100,000) + (100,000 * 10 * 3,000 * 180) = 550,000,000,000$ bytes = 550-GB storage
Estimating network bandwidth to SQL	$(\text{NumSystems} * \text{EstimatedAveBytesPerASC} * \text{Asc/Hr} * \text{ASCtoSQLExpansionFactor} * \text{SystemOnHoursPerDay}) + (\text{NumSystems} * \text{EventsPerDayPerSystem} * \text{AveSizeOfEvent} * \text{EventtoSQLExpansionFactor}) / (\text{HoursPerDay} * \text{MinutesPerHour} * \text{SecondsPerMinute})$	$(100,000 * 23,000 * 1 * 2.5 * 10) + (100,000 * 10 * 3,000 * 1.1) / (24 * 60 * 60) = 703,704$ bytes/sec

 Note

We have not included a generic calculation for IOPS (input/output operations per second) because of the number of variables in a ePO - On-prem environment. Each environment is unique, and includes variables such as administrator query use, SQL index fragmentation, and scheduled reports.

Default numbers for estimating network bandwidth to SQL

Variable	Type	Default
NumSystems		100,000
EstimatedAveBytesPerASC	Bytes	23,000
ASC/Hr	Count (6 hr ASC would be .166)	1
SystemsPerAddlCoreEPOServer		10,000
NumSystemsPerCoreAgentHandler		12,500
ASCtoSQLExpansionFactor		2.5
SystemOnHoursPerDay		10
EventsPerDayPerSystem		10
AveSizeOfEvent	Bytes	3,000
EventtoSQLExpansionFactor		1.1
HoursPerDay	H/D	24
MinutesPerHour	M/H	60

Variable	Type	Default
SecondsPerMinute	S/M	60
BytesPerBlock	B/B	4,000

Sizing distributed repositories

Sizing for distributed repositories depends on the network architecture of the environment.

Distributed repository sizing recommendations:

- Server class hardware or equivalent VM, 2–4 CPU
- Gigabit or greater network interface
- 100-GB disk free space
- Up to 5,000 systems with high latency connections
- Up to 20,000 systems with low latency connections



Tip

When configuring updates, randomize update times across endpoints. Peer-to-peer enabled agents can increase distributed repository system capability by 10–100X, enabling endpoints to share repository files within the network segment.

Example 1: Endpoints concentrated in a single low-latency data center

Fewer than 5,000 endpoints, updating can be done directly from the ePO - On-prem server or Agent Handlers. After the first 5,000 endpoints, start adding a distributed repository for every 20,000 systems.

Actual endpoint count per repository is a factor of client update speed versus concurrent repository connections versus network interface usage.



Tip

Peer-to-peer updating significantly reduces network usage and concurrent connections experienced by the distributed repository, allowing more endpoints per repository.

Example 2: Endpoints concentrated in high-latency data centers

With geographically distant data centers, to reduce WAN bandwidth, place distributed repositories in each zone where high concentrations of endpoints exist. Use the same calculation of one repository for every 20,000 systems.

Example 3: Endpoints concentrated in branch offices with fewer than 1,000 endpoints

Peer-to-peer updating is highly effective in this environment. With peer-to-peer enabled, choose a single SuperAgent distributed repository or no repository at all (preferred). If a local distributed repository is not available, only the initial catalog downloads and peer-to-peer traffic cross the WAN to download from a remote distributed repository.

For information about creating and using SuperAgent repositories, see the product guide for ePO - On-prem.

Example 4: Home office or endpoints directly connected to the Internet

For small endpoint counts, updating can occur using an Agent Handler in a DMZ. For environments where more than 5,000 systems connect to the DMZ, add distributed repositories. These systems typically use connections that are geographically distant. The total concurrent connection load on the distributed repositories is high so the scale factor is lower.

Start with 5,000 systems per distributed repository. Peer-to-peer updating is ineffective because the endpoint's subnet likely has few endpoints (if any) to share content with.

Sizing DXL Brokers

DXL Brokers route messages between clients connected to the DXL messaging fabric. Connect brokers to allow for redundancy, scalability, and communication across different geographical locations.

We recommend a maximum of 50,000 clients per DXL Broker. Size brokers so that if a broker is down, other regional brokers have sufficient capacity to take the load.

For information about redundancy and sizing DXL Brokers, see the DXL Architecture Guide here:

<https://community.trellix.com/t5/Documents/DXL-Architecture-Guide/ta-p/550280?%3E>.

The use cases for DXL Brokers are similar to the ePO - On-prem distributed repositories.

Standalone broker requirements

	Linux	Windows
Recommended requirements		
Processor (CPU cores)	4 cores	4 cores
Memory	8 GB	12 GB
Disk space	25 GB	20 GB
Minimum requirements		
Processor (CPU cores)	2 cores	2 cores
Memory	4 GB	8 GB
Disk space	20 GB	20 GB

Combining servers

Depending on the size of the environment, combining certain services reduces the total number of management servers required for the ePO - On-prem environment.

Depending on the size of your environment, running multiple services on the same physical server require increased CPU, RAM, disk, and network resources.

Use this table as a starting point. Your calculations will depend on your environment.

Total node count	Services being added	Size
< 10,000	ePO - On-prem + DXL Broker	6 cores, 16-GB RAM, 350-GB disk
< 10,000	ePO - On-prem + DXL Broker + SQL	10 cores, 24-GB RAM, 1.5-TB disk
10,000–25,000	ePO - On-prem + DXL Broker	Not recommended

Total node count	Services being added	Size
10,000–25,000	Agent Handler + DXL Broker	8 cores, 16-GB RAM, 200-GB disk
25,000–75,000	ePO - On-prem + DXL Broker	Not recommended
25,000–75,000	Agent Handler + DXL Broker	8 cores, 16-GB RAM, 200-GB disk
75,000–150,000	Agent Handler + DXL Broker	8 cores, 16-GB RAM, 200-GB disk
150,000 +	Agent Handler + DXL Broker	8 cores, 16-GB RAM, 200-GB disk
Other	SuperAgent + DXL Broker < 10,000 systems per server	8 cores, 16-GB RAM, 120-GB disk

- When sharing SQL resources with another service, you must configure SQL Server to only use the resources it would be allocated if run as a separate server. Otherwise, the SQL Server will consume all available system memory and CPU capacity for itself.
- For large environments with 75,000 or more systems, don't share resources on the ePO - On-prem server to maximize the performance of the ePO - On-prem console.
- Combining DXL and repository resources, such as a SuperAgent, lowers the total number of systems that can be served by the shared resource. Each service requires many network connections.

Factors that affect ePO - On-prem performance

It's important to know which factors affect the performance of your server and the attached SQL database.

For example, a single ePO - On-prem server and database can manage up to 200,000 client systems with only the Trellix ENS product installed. But as you add more software products and clients, that same server hardware can no longer provide the performance you expect.

Consider these factors as your managed network grows and your security needs change.

- **ePO - On-prem server hardware**
- **SQL Server** — This server is the main engine within the ePO - On-prem infrastructure and affects the performance of the ePO - On-prem server, queries, dashboards, and ePO - On-prem console.

- **Number of software products installed** — Each software product you install adds processing load on the ePO - On-prem server and the SQL database.
- **Number of managed clients and their Agent Handlers** — These numbers are proportional to the ePO - On-prem server and database performance. Each Agent Handler places these fixed loads on the database server:
 - Heartbeat updates (every minute)
 - Work queue checks (every 10 seconds)
 - Pool of database connections held open to the database (two connections per CPU to the Event Parser service and four connections per CPU to the Apache service)

Internet protocols in a managed environment

ePO - On-prem software is compatible with Internet Protocol versions: IPv4 and IPv6.

The ePO - On-prem server work in three different modes:

- **Only IPv4** — Supports only IPv4 address format
- **Only IPv6** — Supports only IPv6 address format
- **Mixed mode** — Supports IPv4 and IPv6 address formats

The mode in which your ePO - On-prem server works depends on your network configuration. For example, if your network is configured to use only IPv4 addresses, your server works in Only IPv4 mode. Similarly, if your network is configured to use IPv4 and IPv6 addresses, your server works in Mixed mode.

Until IPv6 is installed and enabled, your ePO - On-prem server listens only to IPv4 addresses. When IPv6 is enabled, it works in the mode in which it is configured.

When the ePO - On-prem server communicates with an Agent Handler on IPv6, address-related properties such as IP address, subnet address, and subnet mask are reported in IPv6 format. When transmitted between client and ePO - On-prem server, or when displayed in the user interface or log file, IPv6-related properties are displayed in the expanded form and are enclosed in brackets.

For example, 3FFE:85B:1F1F::A9:1234 is displayed as:

```
[3FFE:085B:1F1F:0000:0000:0000:00A9:1234]
```

When setting an IPv6 address for FTP or HTTP sources, no changes to the address are needed. But, when setting a Literal IPv6 address for a UNC source, you must use the Microsoft Literal IPv6 format. See Microsoft documentation for more information.

 Note

TLS 1.0 is disabled by default for communication to external servers, such as SQL Server. For more information about TLS support, see [KB90222](#). This version of ePO - On-prem requires enabling TLS 1.2 support on your browser.

Things to know before installation

To make sure that you have the necessary information before installing ePO - On-prem, review the checklist.

- Product License Key (not required for evaluations).
- Microsoft SQL authentication requires one of these credentials:
 - Windows authentication credentials — Domain credentials that have Database Owner (dbo) rights on the SQL Server.
 - SQL authentication credentials.
- Destination folder for ePO - On-prem software installation (required for Custom and Cluster installations).
- Installed SQL Server — Provide these details (depending on your configuration) on the Database Information page:
 - The SQL Server name or the SQL Server name with instance name.
 - The dynamic port number used by your SQL Server.
- If you intend to restore the ePO - On-prem server from a database snapshot, you must:
 - Have previously restored the ePO - On-prem SQL database using one of the Microsoft SQL restore processes.
 - Know the server recovery passphrase used with your Disaster Recovery Snapshot records. This passphrase is used to decrypt the sensitive information stored in the SQL Snapshot records.

System requirements and recommendations

Make sure that your environment conforms to all requirements and recommendations before installing ePO - On-prem software.

Run the Pre-Installation Auditor to make sure that your environment meets the minimum requirements for a successful installation. For information about downloading and using the Pre-Installation Auditor, see the tool's release notes.

Component	Requirements and recommendations
Dedicated server	If managing fewer than 250 systems, ePO - On-prem can be installed on a pre-existing server, such as a file server. If managing more than 250 systems, use a dedicated server for ePO - On-prem.
Domain controllers	(Recommended) The server must have a trust relationship with the Domain Controller on the network. For instructions, see the Microsoft product documentation.  Note: Installing the software on a Domain Controller is supported, but not recommended.
File system	NT file system (NTFS) partition.
Free disk space	20 GB — Minimum.
IP address	Use static IP addresses for ePO - On-prem. Static IP addresses are recommended for ePO - On-prem and Agent Handlers. ePO - On-prem supports IPv4 and IPv6 networks.
Memory	8-GB available RAM minimum.
Network Interface Card (NIC)	100 megabit minimum.

Component	Requirements and recommendations
	 Tip If using a server with more than one IP address, ePO - On-prem uses the first identified IP address. To use more IP addresses for agent-server communication, create Agent Handler groups for each IP address. For more information, see KB56281.
Ports	• Make sure that the ports you choose are not already in use on the server system. • Notify network staff of the ports you intend to use for ePO - On-prem and Trellix Agent communication.
Processor	• 64-bit Intel compatible • (Recommended) 4 cores minimum

Software requirements and recommendations

Make sure that you have the required and recommended software installed on your server system before installing ePO - On-prem.

Software	Requirements and recommendations
Microsoft updates	Recommended — Make sure that your Microsoft Windows and Microsoft applications are running the latest updates.

Software	Requirements and recommendations
	 Important Turn off Windows updates before you begin installing or upgrading your software.
Microsoft Visual C++ 2010 Redistributable Package (x64 and x86)	ePO 5.10 Service Pack 1 Update 1 and later does not require this library. The installer includes it only for backward compatibility with specific product extensions.
Microsoft Visual C++ 2013 Redistributable Package (x64 and x86)	ePO 5.10 Service Pack 1 Update 6 and later does not require this library. The installer and CU updater include it only for backward compatibility with specific product extensions.
Microsoft Visual C++ 2015 Redistributable Package (x64 and x86)	Required — Installed automatically.
MSXML 3.0 and 6.0	Required — Installed automatically.
Security software	Recommended. - Install and update the anti-virus software on the server and scan for viruses prior to any installation. - Install and update firewall software on the server.
Supported browser	Recommended — Although it is not a prerequisite for installation, ePO - On-prem requires the use of a supported browser.
Supported SQL Server	Required — A supported version of SQL Server or SQL Server Express is required to install ePO - On-prem.

Software	Requirements and recommendations
SQL Server 2012 (or later) Native Client	Required — Installed automatically.  Note: The automatic installation of the Native Client might fail if the SQL Server is installed on the same system as the ePO server. If this occurs, see KB90874 for more details on how to manually install the client.

Operating system requirements

You can install ePO - On-prem on any supported Microsoft Windows server-class operating system.

Supported server-class operating systems

The software requires one of these supported 64-bit server-class operating systems.

- Windows Server 2012
- Windows Server 2012 Service Pack 1
- Windows Server 2012 R2
- Windows Server 2016
- Windows Server 2019
- Windows Server 2022

For more information on the supported platform, see [KB51569](#).

Important

If you are using Windows Server 2012 or later, also install Microsoft update 2919355.

Operating systems for evaluation

You can use these operating systems to evaluate the ePO - On-prem software, but support is not provided for these operating systems.

- Windows 7 (x64 only)
- Windows 8 and 8.1 (x64 only)
- Windows 10 (x64 only)

Operating system language

ePO - On-prem software runs on any supported operating system regardless of the language of the operating system.

The ePO - On-prem interface has been translated into the languages in this list. When the software is installed on an operating system using a language that is not on this list, the interface tries to display text in English.

- English
- Italian
- English (United Kingdom)
- Chinese (Simplified)
- Chinese (Traditional)
- French
- German
- Japanese
- Korean
- Portuguese (Brazilian)
- Russian
- Spanish
- Turkish

Supported virtual infrastructure software

ePO - On-prem software supports use of several types of virtual infrastructure software.

Supported virtual infrastructure software includes:

- Microsoft Hyper-V Server 2016
- Microsoft Hyper-V Server 2012 R2
- Microsoft Hyper-V Server 2012
- Microsoft Hyper-V Server 2008 R2
- VMware ESXi 6
- VMware ESXi 5.5
- VMware ESXi 5.1
- XenServer 6.2
- XenServer 6

For information about the latest supported platforms, environments, and operating systems for ePO - On-prem, see [KB51569](#).

Supported SQL Servers

ePO - On-prem software requires the use of a supported SQL Server.

ePO - On-prem supports any edition of these Microsoft SQL Server versions.

- Microsoft SQL Server 2014
- Microsoft SQL Server 2016
- Microsoft SQL Server 2017
- Microsoft SQL Server 2019
- Microsoft SQL Server 2022

- Microsoft SQL Server 2025

 Note

ePO - On-prem also supports Amazon RDS for SQL Server which allows you to deploy SQL Server in the cloud.
<https://aws.amazon.com/rds/sqlserver/>

Required SQL Server configuration settings

ePO - On-prem software requires some specific SQL Server configuration settings. For information about working with these settings, see your SQL Server documentation.

Configuration	Details
Nested triggers	The SQL Server Nested Triggers option must be enabled.
Database collation	ePO - On-prem software supports all Microsoft SQL Server collations using the following two SQL collation properties: • Case Insensitivity (CI) • Full ASCII character set support (these characters are included in all Unicode-based character sets) To view the supported Microsoft SQL Server collation types, see KB73717.
Maintenance settings	We recommend making specific maintenance settings to ePO - On-prem databases. For instructions, see the product guide for ePO - On-prem.

Disable Backup Retention Period in Amazon Relational Database Service for SQL Server

If you are using Amazon Relational Database Service (RDS) to manage your SQL Server instance in the cloud, make sure to disable database backups during ePO - On-prem installation.

Complete this task before beginning a new installation of ePO - On-prem.

For more information about working with backups, see the Amazon Relational Database Service (RDS) User Guide: https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_WorkingWithAutomatedBackups.html.

1. Sign into the AWS Management Console and open the RDS console.

<https://console.aws.amazon.com/rds/>

2. In the navigation pane, select Databases, and select the DB instance that you want to modify.
3. Select Modify.

The Modify DB Instance page appears.

4. For Backup Retention Period, select 0 days.
5. Select Continue.
6. Select Apply Immediately.
7. On the confirmation page, select Modify DB Instance.

You are now ready to install ePO - On-prem. After installation, you can return to the RDS console and configure the Backup Retention Period as needed.

Configure TCP/IP access to the SQL Server

ePO - On-prem can only communicate with SQL using a TCP/IP connection. Before installing ePO - On-prem, verify that the SQL Server that will host the ePO - On-prem database has TCP/IP enabled.

Note

Make note of the port that SQL is using.

1. To configure TCP/IP protocol for the SQL Server:
 - a. Start SQL Server Configuration Manager.
 - b. In the console pane, expand SQL Server Network Configuration, and select the Protocols item for your SQL instance. For example, if you are using the default MSSQLSERVER instance, select Protocols for MSSQLSERVER.
 - c. In the details pane, locate the entry for TCP/IP and check the Status column. If it's set to Enabled, go to step 2 to determine the port being used.
 - d. If TCP/IP is set to Disabled, double-click TCP/IP to open the TCP/IP Properties window.

- e. Select the Protocol tab, click Enabled, and select Yes.
- f. Click Apply and then OK to close the Warning dialog.

TCP/IP is enabled. You can now restart the service to make sure that your changes take effect.

- g. In the console pane, click SQL Server Services.
- h. In the details pane, right-click the SQL Server service and click Restart.

2. To determine the port being used by SQL:

- a. If needed, start SQL Server Configuration Manager, expand SQL Server Network Configuration, and select the Protocols item for your SQL instance.
- b. Double-click TCP/IP to open the TCP/IP Properties window.
- c. Select the IP Addresses tab.

Make sure Enabled is set to Yes for each active IP address.

- d. Under IPAll, make note of the value for TCP Dynamic Ports.

If there is a value specified, for example 57482, your SQL Server is using dynamic ports. Make a note of the value because this information might be needed later in the installation.

 Note

If you are using dynamic ports, the SQL Browser service must be running on the SQL Server. If the value for TCP Dynamic Ports is blank, then your SQL Server is using a static port and the value for this port will be shown in the TCP Port field.

- 3. If you are using dynamic ports, make a note of the SQL instance name that will host the ePO - On-prem database. If you are using the default instance for SQL, the instance name is MSSQLSERVER.

Supported Internet browsers

ePO - On-prem software requires the use of one of these supported Internet browsers.

- Firefox 45 and later
- Chrome 51 and later
- Safari 10 and later (macOS only, Windows not supported)

- Microsoft Edge

Note

Internet Explorer is no longer supported in ePO - On-prem 5.10.0 SP1 and later.
For the complete list of supported platforms and browser versions, see [Supported platforms for Trellix ePolicy Orchestrator](#).

TLS requirement

ePO - On-prem console now supports TLS 1.3. Ensure your browser supports TLS 1.2 or 1.3.

Using Internet Explorer enhanced security

If you're using Internet Explorer with enhanced security enabled, add the ePO - On-prem server address to your Internet Explorer trusted sites list (formatted as **https://<servername>**). If you don't, Internet Explorer displays an error message when you try to log on to the ePO - On-prem server.

Agent Handler server requirements

You can install the ePO - On-prem Agent Handler software on any supported Microsoft Windows server-class operating system.

The Agent Handler can authenticate to the ePO - On-prem SQL database using domain credentials. If Windows authentication is not possible, the account the Agent Handler uses to authenticate to the database must use SQL authentication. For more information about Windows and SQL authentication, see the Microsoft SQL Server documentation.

The Agent Handler software requires one of these server-class operating systems:

- Windows Server 2016
- Windows Server 2019
- Windows Server 2022
- Windows Server 2025

Important

For more details on current OS support, see [KB51569](#).

SQL Server installation documented in this guide

ePO - On-prem requires the use of a supported SQL Server. The installation scenario described in detail in this guide assumes that you have already installed a supported version of SQL Server or SQL Server Express.

In this scenario, you install the SQL Server manually and then the Setup program installs the ePO - On-prem software. For more information about installing SQL Server, see your SQL Server software documentation.

Important

If ePO - On-prem is installed in a cluster environment, the SQL Server must be separate from ePO - On-prem: it must not be installed on the same cluster as ePO - On-prem.

Other relevant SQL Server installations and upgrades

See the Microsoft documentation provided for information about these installation scenarios:

- Installing SQL Server 2016, 2017, 2019, 2022, or 2025
- Upgrading from SQL Server 2005 or 2008 to supported SQL Server versions
- Upgrading from SQL Server 2005 Express or 2008 Express to supported SQL Server versions

Required SQL permissions

Specific SQL Server roles are required for the account used by ePO - On-prem.

For new ePO - On-prem installation...	Use these server roles
During installation	The user account credentials for Windows or SQL authentication must have these server roles granted on the target SQL Server: • public • dbcreator

For new ePO - On-prem installation...	Use these server roles
	 Note: The dbcreator server role is required for the setup program to create and add the core ePO - On-prem database objects to the target SQL Server during installation. This ePO - On-prem SQL user account is granted the database role permission db_owner for the ePO - On-prem database.
After the database is created	The dbcreator server role can be removed from the ePO - On-prem SQL user.  Note: Revoking the dbcreator server role restricts the user account to only those permissions granted to the db_owner database role on the ePO - On-prem database.

For an upgrade or patch installation...	Use these roles
During installation	The account credentials for Windows or SQL authentication must have these server roles granted on the target SQL Server: • public • dbcreator

Supported SQL database user name and password formats

Review the supported formats when creating ePO - On-prem and SQL database user names and passwords.

All printable characters in the ISO8859-1 characters set are supported, with these exceptions.

Platform	Unsupported password and user name characters
SQL database	• Leading spaces, trailing spaces, or passwords that contain only spaces • Single quotes (') • Double quotes (") • Leading backslashes (\) • Colons in user names (:) • Semicolons in user names (;)

For more information about supported ePO - On-prem user name and password formats, see [KB66286](#).

Port options

The ports used by ePO - On-prem are predefined, and populated by default.

Review this table for details about which port assignments you can modify.

Port	Default value	Can be changed during installation	Can be changed after installation
Agent-server communication port	80	X	
Agent-server communication secure port	443	X	
Agent wake-up communication port	8081	X	X
Agent broadcast communication port	8082	X	X
Console-to-application server communication port	8443	X	
Client-to-server authenticated	8444	X	

Port	Default value	Can be changed during installation	Can be changed after installation
communication port			
SQL Server TCP port	1433	X	

Automatic product installation

During an automatic installation, ePO - On-prem downloads and installs all Trellix products entitled to you by your ePO - On-prem license key.

Caution

Using Automatic Product Installation downloads all available products in the Software Catalog.

Usually, during an automatic installation, you will not see the Automatic Product Installation process run. It starts running when you complete installing ePO - On-prem and is finished before you log on.

If the Automatic Product Installation page appears when you initially log on to ePO - On-prem, an error occurred while downloading or installing your products. For example, if your Internet connection is interrupted. Make a note of the product that failed to install and click Retry to try the product installation again.

To stop the Automatic Product Installation, click Stop. A confirmation dialog box asks you to confirm that you want to use Software Catalog to install your products.

Important

Once you click OK in the Stop Automatic Product Setup confirmation dialog box, you must use the Software Catalog to install your products, or manually install them in the Main Repository. Automatic Product Installation is available only once during your initial installation.

If a product continues to fail during Automatic Product Installation, contact Technical Support, or click OK to exit the Automatic Product Installation page and begin setting up the ePO - On-prem server.

For future product installation status information, open the Software Catalog: Menu → Software → Software Catalog.

Distributed repository requirements

Distributed repositories are used throughout your environment to provide access to content used by your ePO server. Your distributed repositories must conform to the minimum requirements.

Component	Requirements
Free disk space	1 GB minimum (4 GB recommended) on the drive where the repository is stored. The required space depends on the size of the software packages being replicated from the Main Repository.  Note: The disk space requirement for the distributed repositories on systems where agents are designated as SuperAgents is equal to the disk space available for the Main Repository.
Memory	512 MB minimum.
Repository hosts	• HTTP-compliant servers on Microsoft Windows, or Linux operating systems. • Windows, Linux, or Open Enterprise FTP servers. • Windows, Linux, or UNIX Samba UNC shares. • Systems where a SuperAgent is installed.

Supported products and known issues

Review the products that ePO - On-prem supports and known issues before completing your installation.

- Supported products — [KB94079](#)

- Known issues — [KB90382](#)

Installing ePO - On-prem 5.10.0 on a single server

You can install ePO - On-prem 5.10.0 on Windows operating systems. This topic describes the high-level steps of installing ePO on a single server.

1. [Download ePO - On-prem installation packages](#). For more information about the latest build details, see [ePO - On-prem 5.10.0 Release Notes](#).
2. [Review the minimum supported version of server components and prepare your Windows server ready to install ePO \(KB51569\)](#).
3. [Configure TCP/IP access to the SQL Server](#).
4. Run the Pre-Installation Auditor (PIA) tool to validate that your server meets the minimum requirements. For more information about how to install and run the PIA tool, see the [Pre-Installation Auditor tool documentation](#).
5. [Things to know before installation](#).
6. [Run the setup utility on server to install ePO](#).
7. [Set up your ePO - On-prem environment](#).

Download ePO - On-prem 5.10.0 installation packages

To install ePO - On-prem 5.10.0, you must download the installation packages from the Trellix Product Downloads site using a valid grant number.

1. Go to [Trellix Product Downloads](#).
2. Enter the grant number and email address associated with the product, then click Submit.

The My Products page displays information about your licensed products.

3. In the Find Products section under Filters, select Management Solutions for the Category.

A table displays your licensed ePolicy Orchestrator products.

4. Navigate through the table to locate and select Trellix ePolicy Orchestrator - On-premises.

A table displays all product files that are available to download.

5. In the Available Downloads section, under Filters, select Type as Installation.

The table displays information about each downloadable installation package, including the release date, file size, and notes.

6. Select the Install - ePolicy Orchestrator Version 5.10.0 package.
7. Move the downloaded installation packages to a location where you plan to install ePO, then extract the compressed folder.

The installation packages are ready to run.

Things to know before installation

To make sure that you have the necessary information before installing ePO - On-prem, review the checklist.

- Product License Key (not required for evaluations).
- Microsoft SQL authentication requires one of these credentials:
 - Windows authentication credentials — Domain credentials that have Database Owner (dbo) rights on the SQL Server.
 - SQL authentication credentials.
- Destination folder for ePO - On-prem software installation (required for Custom and Cluster installations).
- Installed SQL Server — Provide these details (depending on your configuration) on the Database Information page:
 - The SQL Server name or the SQL Server name with instance name.
 - The dynamic port number used by your SQL Server.
- If you intend to restore the ePO - On-prem server from a database snapshot, you must:
 - Have previously restored the ePO - On-prem SQL database using one of the Microsoft SQL restore processes.
 - Know the server recovery passphrase used with your Disaster Recovery Snapshot records. This passphrase is used to decrypt the sensitive information stored in the SQL Snapshot records.

Install ePO - On-prem on a single server

Installing ePO - On-prem for the first time requires downloading and starting the installation.

1. Log on to the Windows Server system to be used as the ePO - On-prem server.

Use an account with local administrator permissions.

2. Locate the software you downloaded from the Trellix website and extract the files to a temporary location. Right-click Setup.exe and select Run as Administrator.

The executable is available in the downloaded ePO - On-prem installation folder.

 Caution

If you run Setup.exe without first extracting the contents of the .zip file, the installation fails.

The Trellix ePolicy Orchestrator - InstallShield Wizard starts.

- 3. Click Next to continue installation.

Monitor the installation process for a notification to restart your system.

- 4. To select the destination folder, click:
 - Next to install your ePO - On-prem software in the default location (C:\Program Files (x86)\Trellix\Policy Orchestrator\).
 - Change to specify a custom destination location for your ePO - On-prem software. When the Change Current Destination Folder window opens, browse to the destination and create folders as needed, then click OK.

- 5. The installer discovers active SQL Servers and displays the servers in a drop-down list. Select a SQL server.

If the installer doesn't discover any SQL server, Click No and specify the SQL server details manually.

- 6. In the Database Information, specify information of your database, then click Next.
 - a. Specify the Database Server and Database Name.

Database Server	If the installer discovers active SQL Servers, select your server from the drop-down list. If the installer doesn't discover any SQL server, you can specify the SQL server details manually. If you are using dynamic SQL ports, enter the name of the SQL Server and the name of the SQL instance separated by a backslash. For example, if your SQL Server is called SQLServer and you are using the default instance name of MSSQLSERVER, enter SQLServer\MSSQLSERVER .
Database Name	This value is auto-populated with the name of the database. You can enter a more suitable

	name.
--	-------

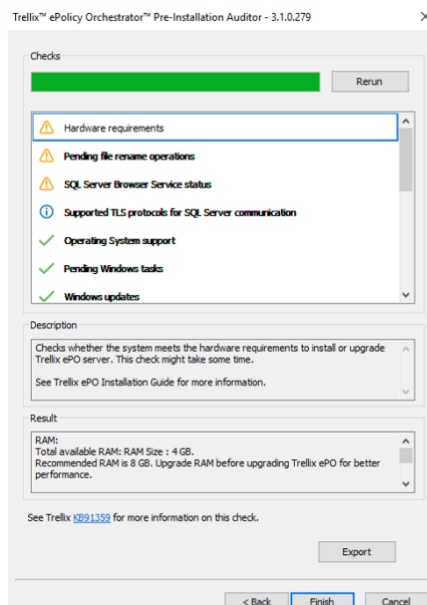
b. Specify which type of Database Server Credentials to use.

Windows authentication	Select a domain of the user account from the drop-down list to access the SQL Server. If the required domain is not listed, type the domain name, user name, and password.
SQL authentication	Type the user name and password for your SQL Server. Make sure that credentials you provide are active and have appropriate administrator rights.  Note: The Domain menu is grayed out once you select the SQL authentication.

c. Click Next.

The installer connects to the SQL Server using the credentials. If the installer can't automatically determine the port, this message appears: Setup was unable to access the SQL UDP port 1434. Click OK to return to the Database Information page.

7. The Pre-Installation Auditor automatically runs and validates the server condition. The checks are passed if the server meets all requirements. If the tool flags warning on a specific issue, you can rectify them and click Rerun. Once all checks have passed, click Finish.



8. In the HTTP Port Information, review the default port assignment, then click Next to verify that the ports are not already in use on this system.

Important

The Agent wake-up communication port and Agent broadcast communication port can only be changed during and after installation. You cannot change the other port numbers after installation. For more information about the port information, see [KB66797](#).

9. In Administrator Information, specify the login credentials and click Next.
 - a. Type the user name and password you want to use for your primary administrator account.
 - b. Type the server recovery passphrase.

The passphrase must be of 14–200 characters, must not contain leading or trailing backslashes (\), spaces, double quotation marks ("), or characters below ASCII 32 or above ASCII 65535.

Important

Save the passphrase as you need it to decrypt the Disaster Recovery Snapshot records. Trellix does not store this passphrase and can't recover it.

Trellix ePolicy Orchestrator 5.10.0 - InstallShield Wizard

Administrator Information
Logon credentials and server recovery passphrase

Administrator credentials used to log on to Trellix ePO

Username:

Password:

Verify Password:

The server recovery passphrase is used to encrypt server snapshots required for recovery. Store it in a safe place; it can't be recovered by Trellix.

Passphrase:

Verify Passphrase:

InstallShield

< Back **Next >** Cancel

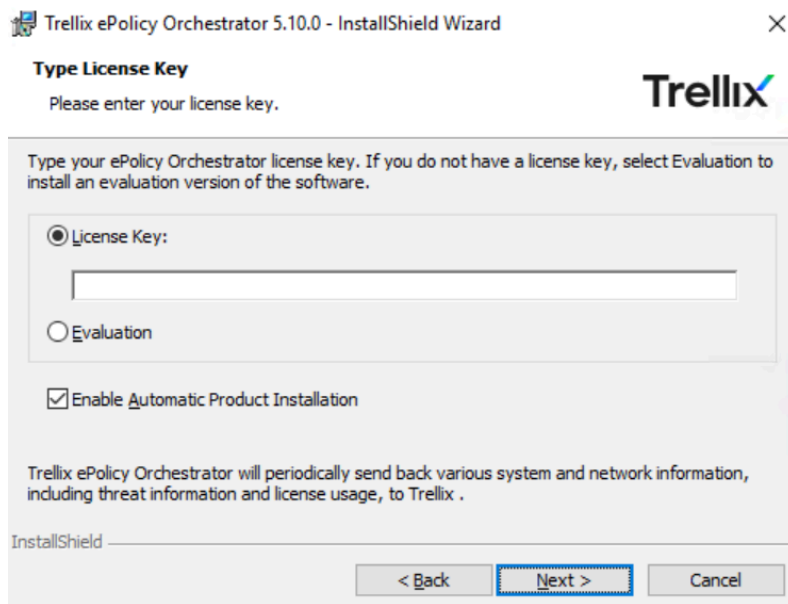
10. In Type License Key, type your license key, then click Next.

If you are evaluating the software and have not purchased the license, select Evaluation to install the software. The evaluation period is limited to 90 days.

1. You can [enable the software license](#) after the installation.
2. (Optional) You can select Enable Automatic Product Installation to automatically download the licensed products after installation.

Note

The Enable Automatic Product Installation option is enabled by default and only available if you have a license key.



11. In Ready to install the Program, you can decide whether to allow Trellix to collect system and software telemetry data, then click Install to begin installing the software.
12. Wait until the installation process is complete, then click Finish to exit the Setup program.

Your server is now installed with ePO - On-prem software.

13. Double-click the Launch ePolicy Orchestrator icon on your desktop to start using your ePO - On-prem server, or browse to the server from a remote web console (**`https://servername:port`**).

Your server is now installed with ePO - On-prem software. Double-click the Launch ePolicy Orchestrator icon on your desktop to start using your ePO - On-prem server, or browse to the server from a remote web console (**`https://servername:port`**).

A certificate warning appears if you are using a self-signed certificate to access ePO - On-prem server through web console. Add the URL to the browser trusted sites.

If you're using Internet Explorer with enhanced security enabled, add the ePO - On-prem server address to your Internet Explorer trusted sites list (formatted as `https://<servername>`). If you don't, Internet Explorer displays an error message when you try to log on to the ePO - On-prem server.

Set recovery model in SQL server

After upgrading to or installing ePO - On-prem 5.10, a new SQL database is created, in addition to the existing ePO - On-prem database, in this format:

```
ePO_<ePOServerName>_Events
```

When you configure SQL database backups, you must add the ePO Events database to your current backup plan.

ePO - On-prem stores monthly event data in the ePO Events database, which handles threat information.

SQL server can be recovered in Full and Simple Recovery models. You can select a recovery model for ePO Events or any database attached to the SQL server.

Full recovery model

- By default, the ePO Events database is configured to use the Full Recovery Model.
- The Full Recovery Model includes the administrative overhead of periodically backing up the Transaction log for your ePO - On-prem database.
- If a disaster recovery is needed, the data and events database must be restored to avoid historical data loss.

Warning

The size of the Transaction log continues to grow if you don't perform periodic backups of the Transaction log when the ePO Events database is in Full Recovery mode. The log grows until it consumes all available disk space, so it is important to perform regular Transaction log backups. Full and differential SQL backups do not truncate the Transaction log.

Simple recovery model

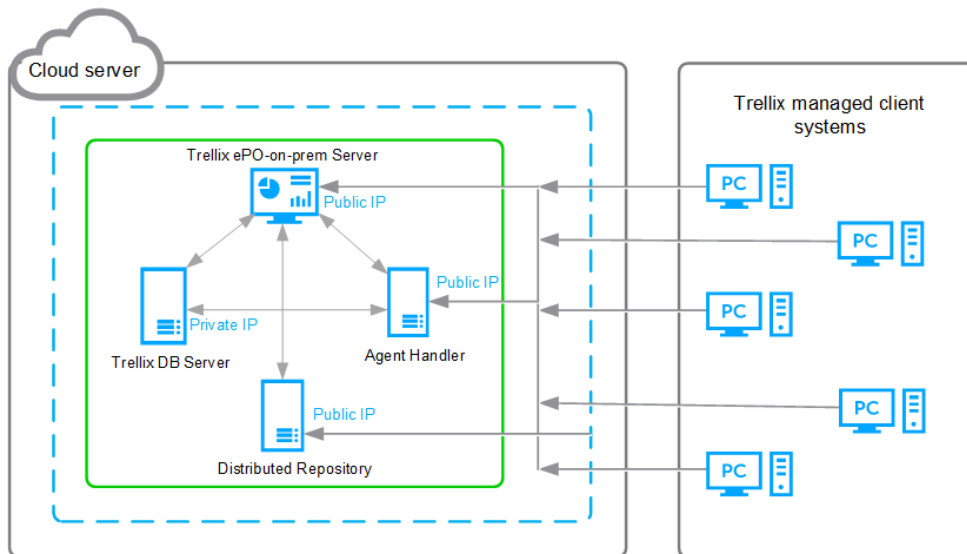
- Set the Simple Recovery mode, if the regular Transaction log backup is not required.
- Simple recovery model performs a full backup of ePO database excluding Transaction logs.
- If there is a disaster, you can recover only to the last full backup. All changes that occurred after the last full backup are lost.
- The Simple Recovery Model is an acceptable solution for most enterprise customers because the data lost in a disaster is usually event data since the last full backup.

Using a Microsoft Azure server for ePO - On-prem

Installing ePO - On-prem on a Microsoft Azure virtual server allows you to resize your server as your network grows, eliminating the chance of hardware failure.

An Azure virtual server provides the same features and performance as locally configured hardware. This diagram shows the basic configuration of ePO - On-prem installed on an Azure server.

Cloud server with ePO - On-prem configuration



Limitations

There are some limitations that you need to consider when a server initiated communication is required.

- If the ePO - On-prem server or the Agent Handler can't communicate with the Agents in a private network, then these features will not work.
 - Push agent doesn't work — Use a VPN to overcome this limitation.
 - Wake up agent using Agent Handler doesn't work — Use a VPN or configure DXL to overcome this limitation.
 - Run client task using an Agent Handler doesn't work — Use a VPN or configure DXL to overcome this limitation.
- If the ePO - On-prem server or the Agent Handler can't communicate with remote servers in private networks, then these features will not work.
 - Distributed repositories such as SuperAgent, FTP, HTTP, and UNC will not work.

- Registered server that cannot communicate with the ePO - On-prem server will not work.
- If ePO - On-prem can't reach the SMTP server, the email service doesn't work.

 Note

If ePO - On-prem can communicate with agents and remote servers, then these features work as expected; provided the required ports are configured in Azure Security Rules.

Port requirements

Configure these ports to establish an uninterrupted communication between ePO - On-prem server, repositories, and the agents.

TCP ports 80 and 443 are the default ports used for communication between ePO - On-prem and the Trellix Agent. You can change the ports while installing ePO - On-prem.

The Azure Inbound Security Rules must allow this communication. For details about port requirements, see [KB66797](#).

In addition to the ports mentioned in the article, the following table lists the ports that you need to configure for these servers:

Server type	Port details
Distributed Server	• Configure port 2049 in the ePO - On-prem server outbound security group. • Configure port 2049 in the distributed repository server inbound security group.
CSR Report Server	• Configure port 9112 in the ePO - On-prem server outbound security group. • Configure port 9111, 9112, 9121 and 9129 in CSR report server inbound security group.
Syslog Server	• Configure port 6514 in the ePO - On-prem server outbound security group. • Configure port 6514 in Syslog server inbound security group.

 Note

Trellix Agent 5.x and later does not support port 80.

Configure the Microsoft Azure server for ePO - On-prem

On the Azure server, you must create a virtual server and start a Virtual Machine (VM) instance to install ePO - On-prem.

You must have a Microsoft Azure account to complete this task.

Perform these steps to install and configure ePO - On-prem in an Azure server to manage your clients.

1. Get an Azure account from <https://azure.microsoft.com/>.
2. Log on to the Azure console and configure your virtual server.
 - a. Start a VM instance.

 Note

Select the location for your virtual server nearest to most of your ePO - On-prem managed systems.

- b. Configure Inbound Security Rules on Azure.

In Azure, a firewall is called as *Inbound Security Rules* and must be created to allow a Trellix Agent to connect to the ePO - On-prem server.

 Note

Make sure that you configure your *Inbound Security Rules* according to ePO - On-prem server port requirements.

- c. Capture the Azure instance public DNS name, or IP address, that Azure created.

 Note

Assign an elastic IP address to the public DNS name or IP address.

3. Use Remote Desktop Connection and the DNS name, or public IP address, to connect to the Azure server.
4. Install ePO - On-prem using software provided by Trellix and information from the Azure SQL database server.
5. Create a Trellix Agent URL or Trellix Agent installation package.

The ePO - On-prem server starts managing your systems.

Install ePO - On-prem on an Azure server

Installing ePO - On-prem on an Azure server is similar to installing the software on a physical server.

- The Azure server must be created.
 - You must know the SQL Server name.
1. Connect to Azure server using Remote Desktop Connection and the configured static IP address or DNS name.
 2. Start the ePO - On-prem installation process.
 3. In Database Information, enter the name of the Microsoft SQL Server configured.

By default, the ePO - On-prem SQL Server name is <AzureServerName>\EPOSERVER.

4. Complete the ePO - On-prem server installation.
5. (Optional) Create a backup image of your Azure server. See Azure documentation for instructions.

You have a ePO - On-prem server installed and configured that you can connect to from a remote browser using this format:

`https://<EPO Server PUBLIC DNS Name>:<port>`

Update ePO - On-prem public DNS name

You must update the ePO - On-prem public DNS name in the console.

ePO - On-prem must be installed on your Azure server.

1. Select Menu → Configuration → Server Settings.
2. Select Trellix ePO Server Public DNS from the Setting Categories pane and click Edit.
3. Enter the Public DNS name and click Save.

Manage your Agent Handlers

You can install an Agent Handler on your Azure server similar to installing an Agent Handler on a physical server.

ePO - On-prem must be installed on your Azure server.

To install and configure an Agent Handler, see the product guide for ePO - On-prem.

- Use an Elastic Load Balancer (ELB) with your Agent Handler to distribute the traffic.
- If an Agent Handler is used without a Load Balancer:

- Go to Menu → Configuration → Agent Handlers.
- Click Agent Handlers under Handler Status.
- Click the Handler DNS Name in the Handler List.
- Enter the Published DNS Name and the Published IP address.
- Click Save.
- If an Agent Handler is used with a Load Balancer:
 - Configure ELB on Microsoft Azure management console.
 - Add Agent Handler VMs.
 - Configure Azure Security Rules for ELB according to port requirements of Agent Handler.
 - For information about how to configure the Load Balancer, see the product guide for ePO - On-prem.

Distributed Repository connections

There are different types of repositories from where the Trellix Agent retrieves the security content to keep the environment up to date.

The packages in the Main Repository are replicated to a distributed repository in the network.

You can create different users for different types of repository and associate them while sharing the folder.

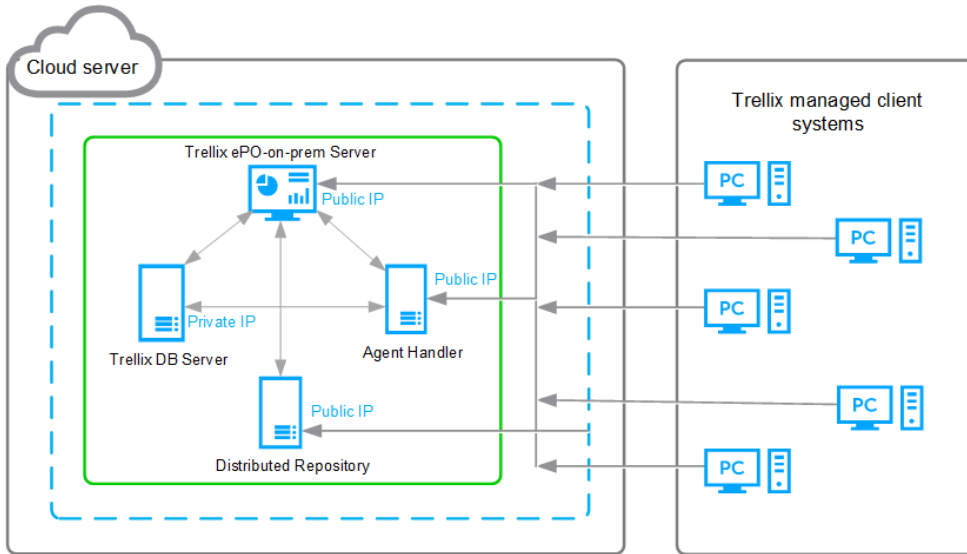
For Universal Naming Convention (UNC) repository, install NFS in the repository server and share the UNC folder using NFS sharing. Open NFS port 2049 in the ePO - On-prem server and in the Repository server.

Using a Google Cloud Platform (GCP) server for ePO - On-prem

Installing ePO - On-prem on a GCP allows you to resize your server as your network grows, eliminating the chance of hardware failure.

A GCP virtual server provides the same features and performance as locally configured hardware. This diagram shows the basic configuration of ePO - On-prem installed on a GCP server.

Cloud server with ePO - On-prem configuration



Limitations

There are some limitations that you need to consider when a server initiated communication is needed.

- If the ePO - On-prem server or the Agent Handler can't communicate with the Agents in a private network, then these features don't work.
 - Push agent doesn't work — Use a VPN to overcome this limitation.
 - Wake up agent using Agent Handler doesn't work — Use a VPN or configure DXL to overcome this limitation.
 - Run client task using an Agent Handler doesn't work — Use a VPN or configure DXL to overcome this limitation.
- If the ePO - On-prem server or the Agent Handler can't communicate with remote servers in private networks, then these features don't work.
 - Distributed repositories such as SuperAgent, FTP, HTTP, and UNC doesn't work.
 - Registered server that can't communicate with the ePO - On-prem server doesn't work.
 - If ePO - On-prem can't reach the SMTP server, the email service doesn't work.

Note

If ePO - On-prem can communicate with agents and remote servers, then these features work as expected; provided the needed ports are configured in Firewall.

Port requirements

Configure these ports to establish an uninterrupted communication between ePO - On-prem server, repositories, and the agents.

TCP ports 80 and 443 are the default ports used for communication between ePO - On-prem and the Trellix Agent. You can change the ports while installing ePO - On-prem.

The GCP firewall rules must allow this communication. For details about port requirements, see [KB66797](#).

In addition to the ports mentioned in the article, the following table lists the ports that you need to configure for these servers:

Server type	Port details
Distributed Server	• Configure port 2049 in the ePO - On-prem server firewall rule. • Configure port 2049 in the distributed repository server inbound security group.
CSR Report Server	• Configure port 9112 in the ePO - On-prem server firewall rule. • Configure port 9111, 9112, 9121 and 9129 in CSR report server inbound security group.
Syslog Server	• Configure port 6514 in the ePO - On-prem server firewall rule. • Configure port 6514 in Syslog server inbound security group.

Note

Trellix Agent 5.x and later does not support port 80.

Configure GCP on ePO - On-prem

On GCP, you must create a virtual server and start a Virtual Machine (VM) instance to install ePO - On-prem.

You must have a GCP account to complete this task.

Perform these steps to install and configure ePO - On-prem in GCP to manage your clients.

1. Get a GCP account from <https://cloud.google.com>.
2. Log on to the GCP console and configure your virtual server.
3. Create a GCP server.
4. In GCP, firewall rules must be created to allow Trellix Agent to connect to the ePO - On-prem server.

 Important

Make sure that you configure your firewall rules according to ePO - On-prem server port requirements.

5. Install ePO - On-prem using software provided by Trellix and information from the SQL database server.

For more information, see the list of [supported SQL servers](#).

The ePO - On-prem server starts managing your systems.

Install ePO - On-prem on a GCP server

Installing ePO - On-prem on a GCP server is similar to installing the software on a physical server.

- The GCP server must be created.
- You must know the SQL Server name.

 Note

If you are using GCP cloud SQL service for ePO - On-prem database, see [KB94509](#) for more information.

1. Connect to the GCP server using Remote Desktop Connection and the configured static IP address or DNS name.
2. Start the ePO - On-prem installation process.
3. In Database Information, enter the name of the Microsoft SQL Server configured.

By default, the ePO - On-prem SQL Server name is <GCPServerName>\EPOSERVER.

4. Complete the ePO - On-prem server installation.
5. (Optional) Create a backup image of your GCP server. See GCP documentation for instructions.

You have a ePO - On-prem server installed and configured that you can connect to from a remote browser using this format:

`https://<GCPServerName>:<port>`

Update ePO - On-prem public DNS name

You must update the ePO - On-prem public DNS name in the console.

ePO - On-prem must be installed on your GCP server.

1. Select Menu → Configuration → Server Settings.
2. Select Trellix ePO Server Public DNS from the Setting Categories pane and click Edit.
3. Enter the Public DNS name and click Save.
4. Run the query, **update [EP0ServerInfo] set RmdSecureHttpPort='443'** on the ePO - On-prem database to enable the secured port 443 for ePO - On-prem DNS.

The ePO - On-prem public DNS name is updated, **`https://<Public DNS NAME>:<port>`**.

Manage your Agent Handlers in the GCP environment

You can install an Agent Handler on your GCP server similar to installing an Agent Handler on a physical server.

ePO - On-prem must be installed on your GCP server.

To install and configure an Agent Handler, see the product guide for ePO - On-prem.

- Use an Elastic Load Balancer (ELB) with your Agent Handler to distribute the traffic.
 - If an Agent Handler is used without a Load Balancer:
 - Go to Menu → Configuration → Agent Handlers.
 - Click Agent Handlers under Handler Status.
 - Click the Handler DNS Name in the Handler List.
 - Enter the Published DNS Name and the Published IP address.
 - Click Save.
 - If an Agent Handler is used with a Load Balancer:
 - Configure ELB on GCP management console.
 - Add Agent Handler VMs.
 - Configure GCP Firewall Rules for ELB according to port requirements of Agent Handler.

For information about how to configure the Load Balancer, see the product guide for ePO - On-prem.

Distributed Repository connections

There are different types of repositories from where the Trellix Agent retrieves the security content to keep the environment up to date.

The packages in the Main Repository are replicated to a distributed repository in the network.

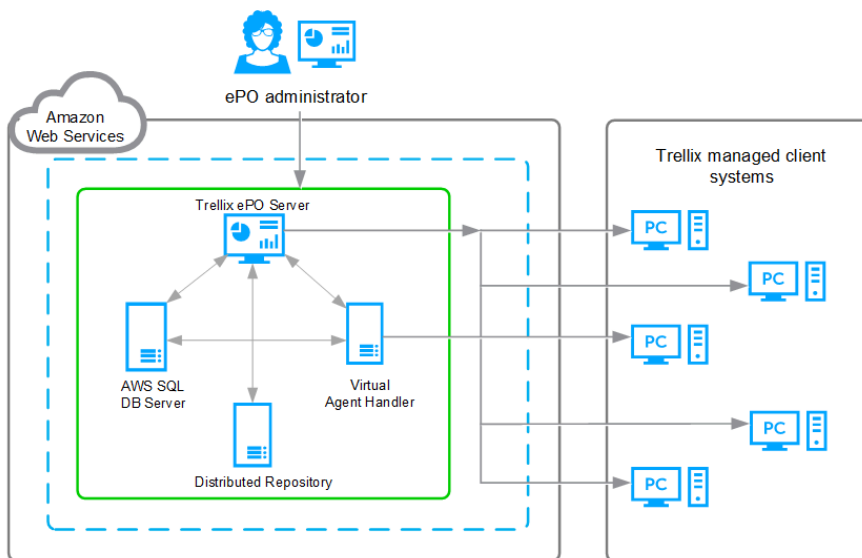
You can create different users for different types of repository and associate them while sharing the folder.

For Universal Naming Convention (UNC) repository, install NFS in the repository server and share the UNC folder using NFS sharing. Open NFS port 2049 in the ePO - On-prem server and in the Repository server.

Installing ePO on AWS server

Installing Trellix ePO on an AWS virtual server lets you easily increase your server size as your network grows and reduces the risk of hardware failures.

AWS virtual servers offer the same features and performance as physical servers. This diagram outlines the basic steps to set up an AWS server to manage your clients with Trellix ePO.



Create an EC2 instance in Virtual Private Cloud

1. Login to AWS console and navigate to EC2 service.
2. Click Launch Instance.

3. Specify the name and click Browse more AMIs.
4. Select the required OS from the displayed list.
5. Select an Instance Type.
6. You can either select a new key pair or an existing key pair.
 - a. Follow these steps to create a new key pair.
 - i. In Key pair (Login), click Create new key pair.
 - ii. Enter a name.
 - iii. Select the key pair type as RSA.
 - iv. Select the Private key file format as .pem.
 - v. Click Create key pair. A file in .pem format is downloaded in your local computer.
 - b. If key pair already exists, select an existing key from the drop-down.
7. Under Network Settings, select VPC-required.
8. Select the subnet that belongs to the created VPC.
9. Select Create security group to create a new security group.

If you have an existing security group, choose Select existing group.
10. Verify the port requirements and open the ports.
11. In Configure Storage, enter the required Storage value.

 Note

20-30 GB is the minimum storage required to install ePO. If you need additional volume, click Add Volume and specify the required volume.

12. Click Launch Instance.

It might take 15-20 minutes for the EC2 instance to connect. You can view the status under Status checks.

- Initializing denotes the EC2 instance creation is in-progress.
- 2/2 checks passed denotes the EC2 instance is ready to connect.

Connect to EC2 instance

You can use key pairs to decrypt and connect to the created AWS EC2 instance.

1. Log on to AWS console.
2. From the console dashboard, choose Launch instances.
3. Right-click the AWS server instance you have created and click Connect to instance.
4. Go to RDP Client → Get Password and launch Get Windows Password.
5. Click Upload private key file and upload the .pem file.
6. Once the RSA private key file loads, Click Decrypt Password.
7. Copy the IP address, username, decrypted password, and the gateway details.

Things to do before installation

1. Create an [AWS account](#).
2. Create a Virtual Private Cloud. For instructions, refer Create a VPC plus other VPC resources in AWS documentation.
3. Create a Bastion Host / Jump server using remote desktop gateway to connect to the created servers. For instructions, refer How to Set Up Remote Desktop Gateway in 4 Steps in V2 Cloud tutorials.
4. [Verify the System and Software requirement to install ePO](#).
5. Select the desired VPC and the private subnet, then create an EC2 instance.
6. Register a domain or use an existing domain. For instructions, refer How to Register a Domain Name with Amazon Route 53 in AWS documentation.
7. Create an Application Load Balancer. For more information, refer AWS documentation.

Install ePO on AWS server

Before you start the installation process, ensure to do the following actions:

- Set up a connection to AWS server.
- Make sure the Trellix ePO server machine and SQL server machine supports communication over TLSv1.1 or TLSv1.2 protocols. For more information, see [KB91823](#).
- Create a Database in EC2 instance with the supported SQL Server.

- Note down the private IP address and the database credentials (sa user) to specify during the installation.
- 1. Launch the Remote Desktop Connection and enter the configured IP address or DNS name of your EC2 instance.
- 2. Enter your credentials and click Connect.
- 3. Start the installation process.
- 4. In Database Information, configure the Microsoft SQL Server using the AWS Server name or the IP address of the database server.
- 5. Create a backup image of your server.
 - a. Log onto AWS Console.
 - b. Click Instances to see a list of your AWS Servers.
 - c. Right-click the AWS Server that you created, then select the Image and Templates.
 - d. Click Create Image. For more instructions, refer AWS documentation.

You have a Trellix ePO - On-prem server installed and configured that you can connect to from a remote browser using this format: `https://<EPO Server PUBLIC DNS Name>:<port>`

Manage your Agent Handlers

You can install an Agent Handler on your AWS server similar to installing an Agent Handler on a physical server.

ePO - On-prem must be installed on your AWS server.

Follow these steps if an Agent Handler is used without a Load Balancer:

1. Go to Menu → Configuration → Agent Handlers.
2. Click New Group and enter virtual Agent Handler group name.
3. In Included Handlers, select Use load balancer and specify Virtual DNS Name and Virtual IP address.
4. Click Save.
5. In Agent Handlers → Handler groups, go to the new virtual AH you have created and Enable it in the Actions column.



Note

Agent Handlers are disabled by default. You need to enable it manually.

6. Click Agent Handlers under Handler Status.

7. Click the Handler DNS Name in the Handler List.
8. Enter the Published DNS Name and the Published IP address.
9. Click Save.
10. Go back to the Agent Handlers page and click New Assignment.
11. Enter a name for assignment.
12. In Agent Criteria, select a system group under System Tree location.
13. In Handler priority, select Use custom Handler and click + to add virtual agent handler.
14. Click Save.

If an Agent Handler is used with a Load Balancer:

1. Configure ELB on AWS console.
 - a. Add Agent Handler VMs.
 - b. Configure Firewall rules for ELB according to port requirements of Agent Handler.

Port requirements

Configure these ports to establish an uninterrupted communication between ePO - On-prem server, repositories, and the agents.

TCP ports 80 and 443 are the default ports used for communication between ePO - On-prem and the Trellix Agent. You can change the ports while installing ePO - On-prem.

The Security groups must allow this communication. For details about port requirements, see [KB66797](#).

In addition to the ports mentioned in the article, the following table lists the ports that you need to configure for these servers:

Server type	Port details
Distributed Server	• Configure port 2049 in the ePO - On-prem server outbound security group. • Configure port 2049 in the distributed repository server inbound security group.

Server type	Port details
CSR Report Server	• Configure port 9112 in the ePO - On-prem server outbound security group. • Configure port 9111, 9112, 9121 and 9129 in CSR report server inbound security group.
Syslog Server	• Configure port 6514 in the ePO - On-prem server outbound security group. • Configure port 6514 in Syslog server inbound security group.

 Note

Trellix Agent 5.x and later does not support port 80.

Update ePO - On-prem public DNS name

You must update the ePO - On-prem public DNS name in the console.

ePO - On-prem must be installed on your AWS server.

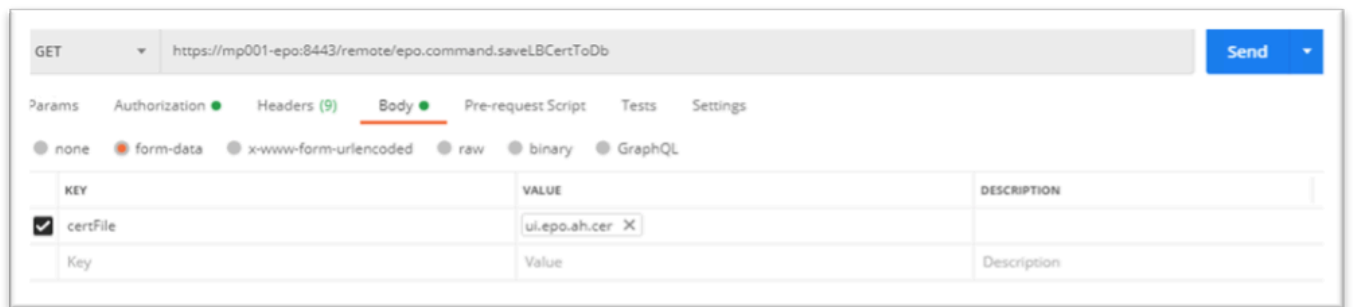
1. Select Menu → Configuration → Server Settings.
2. Select Trellix ePO Server Public DNS from the Setting Categories pane and click Edit.
3. Enter the Public DNS name and click Save.
4. Run the query, **update [EP0ServerInfo] set RmdSecureHttpPort='443', AgentDeploymentURLServer = '<App Server DNS Name>'** on the ePO - On-prem database to enable the secured port 443 for ePO - On-prem DNS.
5. Add Public DNS name in the "EPORegisteredApacheServers" table:
 - a. Update **EPORegisteredApacheServers set PublishedDNSName='<ui-<server name>'** where **DNSName='<ePO Server name>'**
 - b. Update **EPORegisteredApacheServers set PublishedDNSName='ah-<server name>'** where **DNSName='AH server Name'**

The ePO - On-prem public DNS name is updated, **https://<Public DNS NAME>:<port>**.

Importing Certificate

Before you begin, save the ePO and Agent Handler certificates in the PEM format.

1. Enter the Agent handler DNS in your browser. For example, <https://ah-epo.aws.com/>.
2. In browser address bar, click the certificate icon, and click Certificate.
3. Click Global Certificate → View Certificate → Details.
4. Select each certificate and save the certificate in .cer format.
5. Follow the steps 2 to 4 for all certificates.
6. Sign in to Postman and upload the certificates.
 - a. Select each downloaded certificates and select Upload.
 - b. In Authorization tab, enter the ePO credentials to sign in.



Updating host file

Include the IP address and Hostname of the app server and AH server.

1. Open Notepad or any text editor with administrative privileges and navigate to C:\Windows\System32\drivers\etc\hosts.
2. Add the IP address and Hostname of the app server and AH server.
3. Save the changes and close the editor.

Installing ePO - On-prem in a cluster environment

ePO - On-prem provides high availability for server clusters with Microsoft Cluster Server (MSCS) software.

Installing the software into your Microsoft Cluster Server environment requires additional steps. Cluster installation is supported on Windows Server 2008 R2, Windows Server 2012, Windows Server 2016, Windows Server 2019, Windows Server 2022, and Windows Server 2025.

Successful installation depends on proper setup of the Microsoft Cluster Server software. For more information about MSCS setup, see the Microsoft documentation.

Cluster installation terminology

This terminology is used in the cluster installation instructions.

Term	Definition
Data drives	Required by Microsoft Cluster Server and ePO - On-prem. For information about setting up data drives, see the Microsoft documentation. This is the location where you install the ePO - On-prem files.
ePO Virtual IP address resource	The IP address resource that you create as part of the ePO - On-prem cluster installation. This virtual IP address represents the ePO - On-prem cluster installation as a whole. References to this IP address point to the currently active node in your cluster.
ePO Virtual Network Name resource	The Network Name resource that you create as part of the ePO - On-prem cluster installation. This virtual Network Name represents the ePO - On-prem cluster installation as a whole. References to this Network Name point to the currently active node in your cluster.

Cluster installation prerequisites

Before you begin your cluster installation, review this list of requirements and prerequisites, and make sure that each is in place or the information is available. These requirements apply to installations on Windows Server 2008 R2, Windows Server 2012, Windows Server 2016, Windows Server 2019, Windows Server 2022, and Windows Server 2025.

- ePO - On-prem supports two-node cluster environments only. Environments with more than two nodes are not supported.
- Microsoft Cluster Server is set up and running on a cluster of two servers.
- Data drives present and available to all nodes in the cluster according to Microsoft guidelines.
- A supported remote SQL Server is configured.

To confirm that ePO - On-prem can communicate with this server during installation:

- Verify that the SQL Browser Service is running.
 - Make sure that the TCP/IP Protocol is enabled in the SQL Server Configuration Manager.
- You might need to provide these details during the installation process (depending on your configuration), on the Database Information page:
- The name of your SQL Server. Depending on the configuration, format this name using the SQL Server name or the SQL Server name with instance name.
 - The dynamic port number, if any, used by your SQL Server. Specify the dynamic port number during the installation process, on the Database Information page.

Create the ePO - On-prem application role

The ePO - On-prem application role is required to allow Microsoft Cluster Services to control ePO - On-prem.

The steps required to create a ePO - On-prem application role might vary depending on the setup and configuration of your cluster environment. For more information, see the Microsoft documentation.

1. Open the Failover Cluster Manager: click Server Manager → Tools → Failover Cluster Manager.
2. Right-click Roles in the System Tree, then select Create Empty Role.
3. Click OK.
4. Right-click the empty role, then select Properties.
5. In the New Role dialog box, type a name for the role. For example, **ePO**.
6. Click OK.

Create the Client Access Point

The Client Access Point defines the ePO - On-prem Virtual IP address and Virtual Network names so your cluster nodes can communicate with your ePO - On-prem server.

1. Right-click the ePO application role, then select Add a resource → Client Access Point.

The Client Access Point Wizard appears.

2. Type the ePolicy Orchestrator Virtual Name in the Name field and specify the ePolicy Orchestrator Virtual IP address in the Address field, then click Next.

The Confirmation page appears.

3. Click Next to apply the Client Access Point changes, then click Finish when the wizard is complete.
4. If the Client Access Point is offline, right-click the name and select Bring Online.

Add the data drive

The data drive is the location where you install ePO - On-prem.

1. Right-click the ePO application role, then select Add Storage.
2. In the Add Storage dialog box, select the data drive to use for your ePO - On-prem installation, then click OK.

Install ePO - On-prem software on each cluster node

Run the Cluster installation on each of the nodes.

1. Log on to the Windows Server system to be used as the first node of the ePO - On-prem server cluster.

Use an account with local administrator permissions.

2. Locate the software you downloaded from the Trellix website and extract the files to a temporary location. Right-click Setup.exe and select Run as Administrator.

The executable is located in the downloaded ePO - On-prem installation folder.

Caution

If you try to run Setup.exe without first extracting the contents of the .zip file, the installation fails.

The Trellix ePolicy Orchestrator-On-prem- InstallShield Wizard starts. Click Next.

3. On the Setup Type page, select the Cluster option, then click Next.
4. In the Choose Destination Location page, specify the path for the shared data drive, then click Next.

Use the same path for each node.

5. On the first node in the Set Virtual Server Settings page, provide this identifying information for the ePO - On-prem cluster:
 - ePO - On-prem Virtual Server IP address
 - ePO - On-prem Virtual Cluster name
 - ePO - On-prem Virtual Cluster FQDN

On subsequent nodes, the Virtual Server IP address, Virtual Cluster name, and Virtual Cluster FQDN are automatically provided. You must add the Cluster Configuration Passphrase to each subsequent node.

6. The installer searches for SQL Servers. If the installer finds any SQL Servers, it automatically moves to the next stage and the servers that it finds can be selected from a drop-down list.

If it doesn't find any SQL Servers, a dialog box appears asking if you want to search again. Click No to continue to the next step where the SQL Server information can be entered manually.

7. In the Database Information step, specify information for your database, then click Next.
 - a. Specify the Database Server and Database Name.

Database Server	If the installer found the SQL Server in the previous step, select your server from the drop-down list. If the server is not listed, enter the information manually by typing the name of the SQL Server. If you are using dynamic SQL ports, enter the name of the SQL Server and the name of the SQL instance separated by a backslash. For example, if your SQL Server is called SQL Server and you are using the default instance name of MSSQLSERVER, enter SQLServer\MSSQLSERVER.
Database Name	This value is automatically populated with the name of the database. Enter a new database name to change the value.

- b. Specify which type of Database Server Credentials to use.

Windows authentication	From the Domain menu, select the domain of the user account for accessing the SQL Server from the drop-down list. If the required domain is not listed, type the domain name, user name, and password.
SQL authentication	Type the user name and password for your SQL Server. Make sure that credentials you provide represent an existing user on the SQL Server with appropriate rights.  Note: The Domain menu is grayed out when using SQL authentication.

- c. Click Next.

The installer attempts to connect to the SQL Server using the credentials given. If the installer can't automatically determine the port, this message appears: Setup was unable to access the SQL UDP port 1434. Click OK to return to the Database Information page. However, the SQL Server TCP port field is now available. Enter the port and click Next.

8. The Pre-Installation Auditor automatically starts. Review the results and correct any failures, then click Rerun. Once all checks have passed, click Finish.
9. In the HTTP Port Information step, review the default port assignment, then click Next to verify that the ports are not already in use on this system.

Important

You can change some of these ports now. When your installation is complete, you can change only the Agent wake-up communication port and Agent broadcast communication port.

10. In the Administrator Information step, type this information, then click Next.
 - a. Type the user name and password you want to use for your primary administrator account.
 - b. Type the server recovery passphrase.

The passphrase includes 14–200 characters, must not contain leading or trailing backslashes (\), spaces, double quotation marks ("), or characters below ASCII 32 or above ASCII 65535.

Important

Keep a record of this passphrase; you need it to restore ePO - On-prem using the Disaster Recovery Snapshot records and Trellix can't recover it.

11. In the Type License Key step, type your license key, then click Next.

If you don't have a license key, you can select Evaluation to continue installing the software in evaluation mode. The evaluation period is limited to 90 days. You can enter a license key after installation is complete from the ePO - On-prem Settings or Software Catalog. Optionally, if you want ePO - On-prem to automatically download the products you are licensed for after the installation completes, select Enable Automatic Product Installation. For more information, see *Automatic Product Installation*.

Note

The Enable Automatic Product Installation option is enabled by default and only available if you have a license key.

12. Accept the End-User License Agreement and click OK.
13. From the Ready to install the Program dialog box, decide if you want to allow Trellix to collect system and software telemetry data, then click Install to begin installing the software.
14. When the installation is complete, do not select Yes, I wish to launch Trellix ePolicy Orchestrator-On-prem now. Click Finish to exit the Setup program on the first cluster node.
15. In Failover Cluster Manager, move the ePO application role to the second node of the cluster by right-clicking the role, then select Move → Select Node. Select the second node of the cluster and click OK.

The role moves to the second node of the cluster.

Alternatively, shut down the first cluster node server: this automatically moves the role to the second node.

16. Log on to the Windows Server computer to be used as the second node of the ePO - On-prem server cluster.

Use an account with local administrator permissions.

17. Locate the software you downloaded from the Trellix website and extract the files to a temporary location. Right-click Setup.exe and select Run as Administrator.

The executable is located in the downloaded ePO - On-prem installation folder.



If you try to run `Setup.exe` without first extracting the contents of the .zip file, the installation fails.

The Trellix ePolicy Orchestrator-On-prem - InstallShield Wizard starts. Click Next.

18. On the Setup Type page, select the Cluster option, then click Next.
19. In the Choose Destination Location page, click Change, and browse to the location on the shared drive where ePO - On-prem was installed in step 4, then click OK → Next.
20. In the Set Virtual Server Settings step, the details for the ePO - On-prem Virtual Server IP address, ePO - On-prem Virtual Cluster name, and ePO - On-prem Virtual Cluster FQDN are already populated. Enter the Cluster Configuration Passphrase that you chose in step 5 and click Next.
21. From the Ready to install the Program dialog box, decide if you want to allow Trellix to collect system and software telemetry data, then click Install to begin installing the software.

The install process on the second node completes much more quickly than on the first node.

22. When the installation is complete, do not select Yes, I wish to launch Trellix ePolicy Orchestrator-On-prem now. Click Finish to exit the Setup program on the first cluster node.

Create the Generic Service resources

The Generic Service resources enable the cluster server to control the ePO - On-prem server, by starting and stopping the ePO - On-prem services on the correct cluster.

Create three generic service resources.

1. In the Failover Cluster Manager, right-click the ePO application role, then select Add Resource → Generic Service.
2. On the New Resource Wizard, select the ePolicy Orchestrator - On-premises Application Server service, then click Next.
3. On the Confirmation page, click Next to create the service, then click Finish to create the generic service.
4. Repeat steps 1–3 for the ePolicy Orchestrator - On-premises Server service and the ePolicy Orchestrator - On-premises Event Parser service.

The newly created generic service resources are listed on the Resources tab of Failover Cluster Manager under the Roles section. Follow these steps to configure these resources.

5. Right-click the ePolicy Orchestrator - On-premises Application Server resource, then select Properties. In the Properties dialog box, select the Dependencies tab, add the following dependencies, then click Apply → OK.
 - a. Server Name resource

- b. Shared Storage resource
6. Right-click the ePolicy Orchestrator - On-premises Server resource, then select Properties. In the Properties dialog box, remove anything in the Startup Parameters field and enter a single space character. The service will not start with any parameters specified.
7. Select the Dependencies tab, add ePolicy Orchestrator Application Server resource as the dependency, then click Apply → OK.
8. Right-click the ePolicy Orchestrator - On-premises Event Parser resource, then select Properties. In the Properties dialog box, select the Dependencies tab, add the following dependencies, then click Apply → OK.
 - a. Server Name resource
 - b. Shared Storage resource
9. In Failover Cluster Manager, right-click the ePO application role and select Start Role to bring the ePO application role online.

Test the ePO - On-prem cluster installation

When the ePO - On-prem role is online and running in Failover Cluster Manager, use this task to make sure that the software functions in a failover situation.

1. From a separate system, open a web browser and log on to the ePO - On-prem console. The URL for the console is `https://<Server Name Resource>:<console port>`, where <Server Name Resource> is the server name used when the Client Access Point was created, and <console port> is the console port chosen during setup (8443 by default).
2. In Failover Cluster Manager, move the ePO - On-prem application role to the other node of the cluster by right-clicking the role, then select Move → Select Node. Select the other node of the cluster and click OK.

The role moves to the other node of the cluster.

The passive node automatically becomes the active node. The amount of time required for the passive node to become active depends on your unique environment.

3. Manually refresh your browser session. If failover is successful, you are redirected to the ePO - On-prem logon page.

Enabling Multi-Subnet failover support using ePO 5.10.0

Your SQL database with MSSQL-JDBC driver can have both disaster recovery and high availability by enabling the multi-subnet failover support in ePO 5.10.0.

- Server must have ePO 5.10 Service Pack 1 Update 1 installed.
- Check if your extensions are compatible MSSQL-JDBC driver ([KB96549](#)).

1. Launch epo server with administrator permissions.
2. Go to <https://localhost:8443/core/config> in a web browser to launch the Configure Database Settings page.

 Note

8443 is the default port. It may vary based on your server core configuration.

3. In Database Driver, select **MSSQL-JDBC** from the drop-down list.
4. Specify/Check the database server port number.
5. Select the Multisubnet failover checkbox.
6. In Authentication, select **SQL Server Authentication** from the drop-down list.
7. Specify the username and password.
8. Click Test connection and verify the database connection with MSSQL JDBC driver.
9. (Recommended) Restart the ePO service.

For more information on Multi-subnet failover support, refer [KB96625](#).

Setting up your ePO - On-prem 5.10.0 environment

After installing ePO - On-prem 5.10.0, get up-and-running quickly by configuring the essential features of your ePO - On-prem server.

1. Install the Trellix licensed products on the ePO - On-prem server.
 - a. [Automatic Installation](#)
 - b. [Manual Installation](#)
2. [Add your systems to the System Tree](#).
3. [Choose a deployment method to deploy Trellix Agent](#) to your systems to manage them through ePO - On-prem management console.
4. Deploy the Trellix licensed products to your systems.
5. Configure product updates in your managed systems.
6. [Define proxy settings](#), if you use a proxy server in your network environment.
7. [Enable ePO - On-prem 5.10.0 license](#).
8. [Confirm that your systems are managed by ePO - On-prem](#).
9. [Confirm that your protection software stops a sample threat](#). For information about how to use the EICAR anti-malware test with Trellix products, see [KB59742](#).
10. [Confirm the threat response in ePO - On-prem](#).

Install products automatically on your ePO - On-prem server

During an automatic installation, ePO - On-prem downloads and installs all Trellix products approved for your ePO - On-prem license.

- To automatically install, you must have selected Enable Automatic Product Installation during the ePO - On-prem installation.
 - Automatic Product Installation process runs immediately after the ePO installation and finishes before you log on.
 - If installation of all or any product fails, the Automatic Product Installation page appears when you initially log on to ePO - On-prem. You can click Retry to try the product installation again.
1. On your ePO - On-prem server desktop, click the Launch ePolicy Orchestrator icon.

2. When the log-on screen appears, type your credentials and select a default language for the console.

The Product Installation Status software automatically starts downloading and installing the licensed software available to your organization. The page displays this information:

- Products — All licensed software and the latest available version.
- Status — The progress of the product's installation.

3. Wait for the status of each product to change to Complete.
4. To know more about the product installation status information, navigate to Software Catalog in ePO console: Menu → Software → Software Catalog. Search for a product to view the status.

 Note

If any product installation fails, select the checkbox next to the product name to retry installation. If a product repeatedly fails during Automatic Product Installation, contact Technical Support, or click OK to exit the Automatic Product Installation page and begin setting up the ePO - On-prem server.

Install products manually on your ePO - On-prem server

Your Trellix licensed products must first be checked in to the ePO - On-prem server before they can be installed on managed systems.

- If you have selected the Enable Automatic Product Installation, the products are installed along with the ePO installation.
- If you didn't select the Enable Automatic Product Installation option during the ePO - On-prem installation, you can manually check in products on the ePO - On-prem server.

1. On your ePO - On-prem server desktop, click the Launch ePolicy Orchestrator icon.
2. When the Log On screen opens, type your credentials and select a default language for the console.

The default dashboard appears the first time you log on to ePO.

3. Select Menu → Software → Software Catalog.
4. In the Software Catalog page Category list, filter by product categories, or use the search box to find your software.
5. When you have located the correct software, select Check In All.
6. Under Check In, review and accept the product details and End User License Agreement (EULA), select the Client Package Branch, then click Check In to complete the operation.

Add systems to the System Tree

ePO - On-prem uses Agent to communicate with the security products. To deploy Agent initially, you must add systems manually to the System Tree.

- ePO - On-prem server must be connected to the target systems with local administrator access.
- 1. Use **ping** commands to test the ability to successfully resolve and connect from the ePO - On-prem server to managed systems.
- 2. To confirm the **Admin\$** share folder on Windows target systems is accessible from the ePO - On-prem server, click Windows Start → Run, then type the path to the target system's **Admin\$** share, specifying system name or IP address. For example, type `\\<System Name>\Admin$`.

If the systems are properly connected over the network, if your credentials have sufficient rights, and if the Admin\$ share folder is present, a Windows Explorer dialog box appears.

3. Select Menu → Systems → System Tree, then click New Systems on the System Tree page.
4. From the New Systems page, click Push agents and add systems to the current group and Browse.
5. From the NT Domain Credentials dialog box, type this information and click OK.
 - Domain — Type the domain name with your target systems. Use a period (".") to represent a local (non-domain) account.
 - User Name — Type your user name.
 - Password — Type your password.

6. On the Browse for Systems page, select the domain server from the Domain list.

7. Select the systems or groups of systems to add to the System Tree, then click OK.

The systems you selected appear in the Target Systems field, separated by commas.

8. In Agent Version, select Windows or Non-Windows and the version from the list.

9. In Credentials for agent installation:

- Type the domain name.
- Type your domain user name.
- Type the domain password.
- Select the Remember my credentials for future deployments checkbox.

10. Use the defaults for the final settings and click OK.

After Agent successfully establishes communication with ePO the status is set to Managed State. Now you can deploy other Trellix software products on those managed system through ePO. The systems you selected are added to the System Tree and appear as Unmanaged in the Managed State column. This process can take several hours to complete.

Define proxy settings

If you use a proxy server in your network environment, you must specify the proxy settings on the Server Settings page.

1. Select Menu → Configuration → Server Settings, select Proxy Settings from the Setting Categories, then click Edit.
2. Select Configure the proxy settings manually, provide the specific configuration information your proxy server uses for each set of options, then click Save.

Enable ePO - On-prem license

Your license key entitles you to a full installation of ePO 5.10.0, and populates the Software Catalog with the licensed Trellix products your company owns.

Without a license key, ePO runs in evaluation mode. Once the evaluation period expires, ePO ceases to function. You can add a license key at any time during or after the evaluation period.

1. Select Menu → Configuration → Server Settings, select License Key from the Setting Categories, then click Edit.
2. Type your License Key and click Save.

Confirm that your protection software stops a sample threat

Run a sample threat on a managed system to test that your protection software detects and stops it.

Trellix ENS must be installed on the managed system to run the anti-malware test file.

You can also test the software from ePO as they are managed. In this example we consider the software is ENS and test its successful function.

To run an anti-malware test file, you can log on to the test system locally or remotely.

1. Log on to the test system with administrator rights.
2. Using a web browser, connect to the EICAR site:

<http://www.eicar.org/86-0-Intended-use.html>
3. Follow the instructions to download and run the 68-Byte eicar.com anti-malware test file.
4. In Windows, click Start → All Programs → Trellix → Trellix Endpoint Security, then click Status.

A threat Summary lists the type of threat and the number of threats received.

5. Click Event Log to display the threat events in the Event table and remove the sample threat.

The bottom pane of the Event table lists the threat event details.

Confirm the threat response in ePO - On-prem

Knowing where to look for threat events in ePO - On-prem is an important part of protecting your managed systems.

You must run the EICAR anti-malware test file to see any threat events.

1. Log on to the ePO - On-prem and select Menu → Reporting → Dashboards.
2. In the title bar of the Dashboards list, select a dashboard.
 - ePO Summary — Displays recent threats in the Malware Detection History line chart.
 - Executive Dashboard — Displays a Malware Detection History line chart.
 - Threat Events — Displays recent threats in these dashboards:
 - Most Numerous Threat Event Descriptions table
 - Threat Events by System Tree Group table and pie chart
 - Threat Event Descriptions in the Last 24 Hours table and pie chart
 - Threat Events in the Last 2 Weeks line chart
3. Select Menu → Reporting → Threat Event Log to see a description the recent threat.

Information in the table includes:

- Event Generated Time
 - Event ID
 - Event Description
 - Event Category
 - Threat Target IPv4 Address
 - Action Taken
 - Threat Type
4. Click the event in the table to see all threat details.

5. To see the detailed information about the affected system, click Go to related Systems.

Confirm that your systems are being managed by ePO

After deploying Trellix Agent and the licensed products, make sure that your systems are listed in the System Tree and appear as managed.

You must have deployed the Trellix Agent and downloaded the product software to your systems.

1. Select Menu → Systems → System Tree, then click the Systems tab to show a list of managed systems.



If no systems appear, click This Group and All Subgroups in the Preset list.

2. In the Managed State column, confirm that Managed appears next to each system.

If Unmanaged appears in the Managed State column, the system was added to the System Tree but the Trellix Agent and product software are not installed on the system.

3. To view details about a system, select the system name to open the Systems Information page.

Upgrade to ePO - On-prem 5.10.0 Service Pack 1

Upgrading your existing ePO - On-prem environment involves careful planning and preparation for a smooth and successful process.

The ePO - On-prem upgrade process consists of three phases:

Phase 1: Before you upgrade — Check that your ePO - On-prem server meets all requirements, download the ePO - On-prem software and tools, and back up your databases and directories.

Important

Make sure to review the supported upgrade paths. For more details, see [KB96141](#).

If your current ePO version is...	Upgrade to...	Cannot upgrade to...
ePO 5.3.x ePO 5.9.x	Trellix ePO 5.10 Service Pack 1	-
ePO 5.10.0 (CU1 to CU15)	Trellix ePO 5.10.0 Service Pack 1 Update	Trellix ePO 5.10 Service Pack 1 or later
ePO 5.10.0 (2428.68)	Trellix ePO 5.10.0 Service Pack 1 Update	Trellix ePO 5.10 Service Pack 1 or later
Trellix ePO 5.10.0 Service Pack 1 Update	-	Trellix ePO 5.10 Service Pack 1 or later
Trellix ePO 5.10 Service Pack 1	Trellix ePO 5.10.0 Service Pack 1 Update 1	-

Phase 2: Day of upgrade — Stop Agent Handler and ePO - On-prem services, start and complete the InstallShield Wizard, and

upgrade Agent Handlers.

Phase 3: After you upgrade — Complete the post-upgrade checklist and migrate SHA-1 certificates to SHA-2 or higher.

Pre-upgrade checklist

The pre-upgrade checklist identifies what to do before you begin your upgrade, so you have a successful upgrade and limited downtime.

1. Plan for a successful upgrade.
 - [Review the supported upgrade paths \(KB96141\)](#).
 - Download the Pre-Installation Auditor.
 - Download ePO - On-prem software.
 - Run the Pre-Installation Auditor to check for issues that could cause the upgrade to fail, and compatibility of installed products.
2. Prepare your environment.
 - Back up ePO - On-prem databases and directories.
 - Update database server certificates.

Mandatory prerequisites and upgrade requirements

Important details about Service pack 1 Updates

Before you upgrade

For more details on upgrade path scenarios, see [KB96141](#).

- Ensure to back up the existing ePO - On-prem 5.10.0 setup.
- Ensure to migrate SHA-1 to SHA-2. For instructions on how to migrate, see [KB87017](#).
- Windows Server 2012 and SQL Server 2012 are not supported from this release. For complete system requirements, see supported platforms for ePO - On-prem 5.10.0 ([KB51569](#)).
- Applicable for update 5 and 6: Upgrade DXL Brokers and Extensions to 6.1.1

 Note

If the user wants to install/upgrade OnPrem ePO SP1 Update 5/6, it is mandatory to upgrade DXL Brokers and Extensions to 6.1.1. It is recommended to upgrade all the DXL Brokers to 6.1.1 before upgrading DXL Extensions to 6.1.1.

After upgrading the 6.1.1 DXL Brokers and Extensions if any connectivity issues encountered please restart the ePO services. For more information, see [KB14602](#).

Planning your ePO - On-prem upgrade

Before you start the ePO - On-prem upgrade, make sure that you have the information you need.

Review documentation

Read the release notes for your version of ePO - On-prem, particularly these sections:

- Supported upgrade paths (See [KB96141](#))
- Supported environments (See [KB51569](#))
- Known issues (See [KB90382](#))

Gather required information

- Grant number

 Note

If you need help with your grant number, contact Trellix support.

- Database server and database name
- Database server credentials, including domain if authenticating with Windows credentials
- Primary ePO - On-prem administrator account credentials
- Disaster Recovery passphrase

Download the tools and software you need to upgrade

Before you upgrade ePO - On-prem, download the Pre-Installation Auditor and ePO - On-prem installation software packages from the Product Downloads site.

1. Go to the [Product Downloads](#) site.
2. Type your grant number and email address, and click Submit.

 Note

If you need help with your grant number, contact Customer Service.

3. On the My Products page, click Trellix ePolicy Orchestrator.

Make sure Show only latest version is selected.

4. On the Available Downloads page:
 - a. Click Install - Trellix ePolicy Orchestrator Version 5.10.0 Service Pack 1.
 - b. Click Trellix ePO - Pre-Installation Auditor tool.
 - c. Click Trellix ePolicy Orchestrator Version 5.10.0 Service Pack 1 Update #.

 Note

The number corresponds with the latest available update. The update is needed when the ePO - On-prem upgrade successfully completes.

5. Extract the downloaded .zip to a temporary location.

 Caution

If you try to run Setup.exe before extracting the content of the .zip file, the installation fails.

Checking your system for compatibility using the PIA tool

Before you install or upgrade ePO - On-prem, run the Pre-Installation Auditor (PIA) tool to reduce or prevent upgrade issues. Running the tool automates the verification tasks included in the upgrade process.

Running a Product Compatibility check

The PIA tool runs a Product Compatibility Check to verify that your managed products are compatible with the latest version of ePO - On-prem. If it finds discrepancies, the tool creates a list of blocked or disabled product extensions.

- Blocked extensions — Upgrade of ePO - On-prem is blocked.

- Disabled extensions — Upgrade continues, but extension isn't initialized until a compatible replacement is installed.

For a complete list of the checks the PIA performs and information about how to run it at any time, see [KB71825](#).

Upload Product Compatibility List (PCL) file manually

You can now provide your own product compatibility list to know whether the product extensions are compatible with the ePO - On-prem version that you are trying to install.

1. Copy the PCL file to ePO server desktop.



Do not change the file name. The file must be named ProductCompatibilityList.xml.

2. Log on to the ePO console.
3. Go to Menu → Configuration → Server Settings.
4. Click Product Compatibility List under the Setting Categories column and click Edit.
5. Select Disabled and click Browse.
6. Go to the location where ProductCompatibilityList.xml is copied in your desktop and click Save.

Back up ePO - On-prem databases and directories

Before upgrading ePO - On-prem, back up your databases and directories to make sure you can recover your ePO - On-prem environment if the upgrade is unsuccessful.

Make sure that you have your Disaster Recovery snapshot passphrase. If you don't know it, see [Change the server recovery passphrase](#).

1. Run a successful Disaster Recovery Snapshot server task. For details, see [Disaster Recovery](#).
2. Back up all ePO - On-prem databases and directories. For details, see [KB66616](#).

Update your database server certificates

Make sure that the certificates for any registered servers that ePO - On-prem communicates with are supported by ePO - On-prem.

ePO - On-prem might not be able to connect to registered servers that use less secure certificates, such as certificates with RSA public key lengths of only 1024 bits. For more information, including additional supported public key algorithms and key lengths,

see [KB87731](#).

 Note

TLS 1.0 is disabled by default for communication to external servers, such as SQL Server. For more information about TLS support, see [KB90222](#).

- Use certificates with RSA public key lengths of 2048 bits or greater for the registered servers that ePO - On-prem connects to.

Upgrade checklist

The upgrade checklist identifies actions that must be performed when you upgrade ePO - On-prem.

 Important

You must complete all steps in the pre-upgrade checklist before you upgrade.

Extra steps are required if you installed ePO - On-prem in a cluster environment. For information about cluster environment, see *Upgrade your ePO - On-prem cluster server*.

1. Confirm you have successfully backed up your ePO - On-prem databases and directories.
2. Stop Remote Agent Handlers.
3. Prepare ePO - On-prem services for upgrade.
4. Run the InstallShield Wizard.
5. Upgrade Agent Handlers (if applicable).

Stop Agent Handler and ePO - On-prem services

If you use Agent Handlers in your environment, you must stop Trellix services on each Agent Handler server before the upgrade.

- Complete all steps in the *Pre-upgrade checklist*.
 1. Stop Remote Agent Handler services.
 - a. Press **Windows+R**, type `services.msc`, then click OK.
 - b. Stop the ePolicy Orchestrator Server Service and ePolicy Orchestrator Event Parser Service.
 2. Stop ePO - On-prem services.

- a. Press **Windows+R**, type `services.msc`, then click OK.
- b. Stop these services:
 - ePolicy Orchestrator Server Service
 - ePolicy Orchestrator Event Parser Service
 - ePolicy Orchestrator Application Server
3. Start the ePolicy Orchestrator Application Server service.

Run the InstallShield Wizard

Run the InstallShield Wizard to upgrade your ePO - On-prem server.



Tip

Monitor the upgrade process. You might need to restart your system.

1. Log on to the system using an account with local administrator permissions and find the `Setup.exe` file.

The executable is located in the folder where it was extracted.



Important

If you try to run `Setup.exe` before extracting the contents of the `.zip` file, the installation fails.

2. To start the ePO - On-prem InstallShield wizard, right-click the `Setup.exe` file, and select Run as Administrator.
3. In the Welcome dialog box of the installation wizard, click Next.

A warning message might appear listing products from your previous version of ePO - On-prem that are no longer supported with this version of the software. These products are not migrated to the new ePO - On-prem repository.

4. To install the listed software, click Next.
5. In the Database Information step:
 - a. Confirm that the automatically selected Database Server and Database Name are correct.
 - b. Specify the type of credentials to use from the lists.
6. Specify which type of Database Server Credentials to use, then click Next.

- Windows authentication — From the Domain menu, select the domain of the user account you're going to use to access the SQL Server. Type the User name and Password. If you are using a previously installed SQL Server, make sure that your user account has access.
- SQL authentication — Type the User name and Password for your SQL Server. Make sure that credentials you provide represent an existing user on the SQL Server with appropriate rights.

**Tip**

The Domain menu is grayed out when using SQL authentication.

The ePO - On-prem Pre-Installation Auditor runs, analyzing your ePO - On-prem environment to make sure it meets all requirements.

7. In the Administrator Information step:

- a. For the Username, replace the default `admin` if necessary, and type your primary Administrator account user name.
- b. For the Password, type your primary Administrator account password.
- c. For the Server recovery password, type a passphrase to encrypt Disaster Recovery Snapshot records.

The passphrase includes 14–200 characters, must not contain leading or trailing backslashes (\), spaces, double quotation marks ("), or characters below ASCII 32 or above ASCII 65535.

**Note**

Keep a record of this password. If you ever want to restore ePO - On-prem from a Disaster Recovery snapshot, you need this password to decrypt the Disaster Recovery Snapshot records.

- d. Click Next.

8. In the Type License Key step, click Next.

Your existing license key is automatically populated in the field and the Product Compatibility Check runs.

9. Accept the Trellix End User License Agreement and click OK.

10. In the Ready to Install the Program dialog box, decide if you want to send anonymous usage information to Trellix, then click Install.

11. When the upgrade is complete, click Next, click Finish, then optionally select to restart the server.

 Note

During the upgrade process, if your ePO - On-prem database is large, the process might take a long time and this message appears: Your ePO - On-prem database has too many events. Your upgrade might take a long time. For information about removing old events, see [KB68961](#).

12. In the InstallShield Wizard Completed dialog box, click Finish to complete the installation.

If you want, click Yes, I want to start Trellix ePolicy Orchestrator-On-prem now.

Your ePO - On-prem software is now updated. Double-click the Trellix icon on your desktop to start using your ePO - On-prem server, or browse to the server (**`https://<servername>:<port>`**).

Upgrade your Agent Handlers

When you upgrade your ePO - On-prem server software, upgrade any Agent Handlers installed throughout your environment. Agent Handlers must be upgraded separately.

Agent Handlers installed with previous versions of ePO - On-prem software are not compatible with this new version, and are not upgraded automatically.

1. Copy the Agent Handler folder, included in the ePO - On-prem software installation package, to the system.
2. Right-click Setup.exe and select Run as Administrator to start the ePO - On-prem Agent Handler InstallShield Wizard.
3. Click Next to begin the upgrade process.
4. Accept the license agreement, then click OK.
5. Accept the default destination or select a different destination, then click Next.
6. Configure the ePO - On-prem server information.
 1. Type the system name of the ePO - On-prem server with which the Agent Handler must communicate.
 2. Specify which port to use for Agent Handler-to-server communication. The default port is 8443.
 3. Type the name and password of a user with ePO - On-prem Global Administrator rights, and click Next.
 4. Provide the password for access to the ePO - On-prem SQL database, then click Next. The Database Information page is populated with these ePO - On-prem server settings.
 - Database Server with instance name, as in DB-SERVER\SERVERNAME.
 - Authentication type.
 - Domain name where the database server is hosted if using Windows authentication.
 - User name and Password.

- Database name if not provided automatically.

7. Click Install to start the installation.

The installation continues without any additional input. When the wizard is complete, click Finish.

Upgrade your ePO - On-prem cluster server

Upgrading your ePO - On-prem software in a cluster environment requires special consideration.

Make sure your current environment is supported for the upgrade to ePO - On-prem 5.10.0 Service Pack 1.

1. In Failover Cluster Manager on the active node, open the ePO group.
2. Make sure that the primary node is the active server.
3. Take this Generic Service resource offline:
 - ePolicy Orchestrator - On-premises Application Server
 - ePolicy Orchestrator - On-premises Event Parser service
 - ePolicy Orchestrator - On-premises Server service
4. Do not change these resources, which are required for a successful upgrade:
 - Data drive
 - ePO - On-prem virtual IP address
 - ePO - On-prem virtual Network Name
5. Open the Services Control Manager and start each of these services on the primary node:
 - ePolicy Orchestrator - On-premises Application Server
 - ePolicy Orchestrator - On-premises Event Parser service
 - ePolicy Orchestrator - On-premises Server service
6. Install the new ePO - On-prem software on the primary node.
7. In Failover Cluster Manager, move the ePO application role to the second node of the cluster: right-click the role, select Move → Select Node. Select the second node of the cluster and click OK.

The role moves to the second node of the cluster.

(Optional) Shut down the first cluster node server, which automatically moves the role to the second node.

8. Install your new ePO - On-prem software on the second node.

9. Create the new Generic Service resources.

For information about how to create the Generic Service resources, see [Create the Generic Service resources](#).

Post-upgrade checklist

After the upgrade is complete, perform a quick health check on your ePO - On-prem server.

Note

ePO - On-prem 5.10 can no longer connect to a registered server that requires TLS certificates but does not support TLS 1.1 or 1.2. For more information, see [Update your database server certificates](#).

1. Confirm you can log on and access different areas of the ePO - On-prem console.
2. Update your certificates to SHA-2 if a warning appears on the ePO - On-prem logon screen. For details, see [Migrate Certificate Authority Hashing Algorithm from SHA-1 to SHA-2 or higher](#).
3. Verify connectivity by sending an agent wake-up call to one or more systems.
4. Verify ePO - On-prem is operating correctly by running a query or server task.
5. Make sure that your policies, tasks, product deployments, and repositories reflect the choices you made when you set up your environment.
6. Test the connection to all your registered servers. If any server fails that uses SSL, it might need to be updated to support TLS 1.1 or higher for ePO - On-prem to successfully connect to it.
7. Apply the available update to ePO - On-prem and all Agent Handlers. For details, see [Trellix ePO Cumulative Updates](#).
8. Enable any server tasks you disabled before upgrading. For more details on supported upgrade paths, see [KB96141](#).
9. Enable Windows updates to make sure that your servers receive the latest updates for your operating system.
10. Apply the latest update when the upgrade is complete.

Migrate Certificate Authority Hashing Algorithm from SHA-1 to SHA-2 or higher

To remediate vulnerabilities in your ePO - On-prem environment, migrate your existing certificates to more secure algorithm certificates or regenerate them.

The SHA-1 algorithm has reached end-of-life (EOL). Many organizations are deprecating TLS/SSL certificates signed by the SHA-1 algorithm. If you continue to use SHA-1 certificates, browsers such as Google Chrome or Microsoft Internet Explorer will flag the ePO - On-prem console as an unsecure HTTPS site.

If you have upgraded ePO - On-prem from an older version, migrate ePO - On-prem certificates to the latest hash algorithm. A fresh installation of ePO - On-prem installs the latest hash algorithm certificates.

The Certificate Manager allows you to:

- Migrate certificates that are signed by older signing algorithm to the new algorithm such as SHA-1 to SHA-256.
- Regenerate your certificates when your existing certificates are compromised due to vulnerabilities in your environment.
- Migrate or regenerate certificates for managed products that are derived from ePO - On-prem root CA.

This task replaces certificates that are used for:

- Agent-server communication
- Authenticating to browsers
- Certificate-based user authentication

Important

Read these instructions carefully before proceeding with the steps. If you activate the new certificates before they are populated on the systems in your network, those systems won't be able to connect to your ePO - On-prem server until the agents on those systems are re-installed.

1. Log on as an administrator, then select Menu → Configuration → Certificate Manager.

The Certificate Manager page provides information about the installed Root Certificate, Agent Handler certificates, server certificates, and other certificates that are derived from ePO - On-prem root Certificate Authority (CA).

2. Click Regenerate Certificate, then click OK to confirm.

The ePO - On-prem root CA and other certificates that are derived from the root CA are regenerated and stored in a temporary location on the server. The time required to complete the process depends on the number of Agent Handlers and extensions that derive certificates from ePO - On-prem root CA.

3. After the certificates regenerate, wait for sufficient saturation of the new certificates throughout your environment.

As agents communicate to the ePO - On-prem server, they are given the new certificate. The percentage of agents that have received the newly-generated certificates is provided in the Certificate Manager under Product: Agent Handler → Status.

 Important

Make sure that the distribution percentage is as close to 100% as possible before you continue. Otherwise, pending systems might not receive the newly generated certificates and won't be able to communicate with the ePO - On-prem after the certificates are activated. You can stay in this state for as long as is necessary to achieve sufficient saturation.

4. Once you've achieved a distribution percentage close to 100%, click **Activate Certificates** to carry out all future operations using the new certificates.

A backup of the original certificates is created, and a message appears.

5. Click **OK**.
6. Stop and start these services:

- a. Stop the Agent Handler services.
- b. Restart the ePO - On-prem services.
- c. Start the Agent Handler services.

7. Monitor your environment and make sure that your agents are successfully communicating.

You can cancel the migration at this point to roll back the certificate and restore agent-to-server communication; however, this is not possible after you have completed the next step.

8. Click **Finish Migration** to complete the certificate migration.

For any issues during the migration, click **Cancel Migration** to revert to the previous certificates. If you cancel the migration, stop the Agent Handler services, restart the ePO - On-prem service, and start the Agent Handler service again.

You can start the certificate migration again after fixing any issues.

9. Re-install any agents that use the old certificates to restore agent-server communication.

ePO - On-prem cumulative updates

ePO - On-prem 5.10.0 introduces a new strategy where we release cumulative updates instead of patches. Updates address product defects and often introduce new features. Because updates are cumulative, you only need to apply the latest available update.

Updates are on the product download site and the Software Catalog. When a new update is released, the previous update is removed from the product download site and Software Catalog so only the latest update is available.

Updates are posted with the following naming convention: ePO 5.10.0 Service Pack 1 Update #

Cumulative update installation procedures

Installation instructions

1. [Install the latest update on your ePO - On-prem server](#)
2. [Repair the latest on previously applied Service pack 1 Update build](#)
3. [Install the latest update on Agent Handlers](#)
4. [Verify the update installation](#)
5. [Roll back the updates](#)

Install the latest update on your ePO - On-prem 5.10.0 server

Make sure your system meets these requirements before installing the latest update:

- ePO - On-prem 5.10.0 is installed.
- Administrator permissions are needed.
- Have a recent snapshot and a backup of the ePO - On-prem database and installation directory.
- Have a minimum of 2-GB disk space is available to copy, extract, and apply updates.

Follow these actions before running the updater tool:

- Standalone environment
 - Stop the ePO - On-prem Application Server Service (Tomcat).
 - Stop the Agent Handler Service, including Remote Agent Handler Service (Apache).
 - Stop the Event Parser Service.
 - Close all other File Explorer windows.
- Cluster environment
 - Bring the Generic Service resources offline (Apache, TomCat, and Event Parser).
 - Copy and extract the contents of ePO 5.10.0 Service Pack 1 Update.zip to the data drive.
 - Run this update on one node.

 Note

These steps apply to all Remote Agent Handlers; perform these steps on each Agent Handler in your environment after you install the latest update.

1. Download ePO - On-prem 5.10.0 Service Pack 1 Update package from ePO Software Catalog or from [Trellix product downloads](#) site.
2. Copy and extract the update package to your ePO - On-prem server.
3. Run ePOUpdater.exe from the folder where you extracted the update package.
4. Enter the database password and click Continue.
5. Review the list of fixes in the Current Update(s) to be installed dialog box and click Continue.

 Note

Click Details to view the file types based on version, action being taken, plug-in name (if applicable), and the destination folder where the files are updated.

6. In the Apply Update pop-up window, select the checkbox if you want the updater tool to migrate the *TrellixFtp* source site to point to the HTTP server.

 Note

The service provider that Trellix uses to host our FTP service no longer provides FTP capabilities. Trellix uses HTTP sites to update content for Trellix products. For more information about End of Support for the FTP option to access CommonUpdater download sites, see [KB91260](#).

By default, the option to update *TrellixFtp* source site details are automatically selected. If you continue with this option, the updater tool changes to the *TrellixFtp* source site setting.

These properties are updated:

Property name	Before upgrade	After upgrade
Source Site Name	McAfeeFtp	TrellixFtpMigrated
Source Site ID	McAfeeFtp	TrellixFtpMigrated
Location	ftp.nai.com/CommonUpdater	update.nai.com/Products/ CommonUpdater

Property name	Before upgrade	After upgrade
Protocol	FTP	HTTP
Port	21	80

 Note

If you choose not to update *TrellixFtp* source site details using the updater tool, you can deselect the checkbox and manually update source site details later.

- Click OK to apply the update.

 Note

An error message appears if services do not stop. In the error message, click OK then manually stop the ePO - On-prem services, and click Re-Apply. This option doesn't apply to ePO - On-prem in a Windows Cluster environment.

- Verify that Successful is highlighted in green after the update completes.

For ePO - On-prem in a Windows cluster, make sure that the generic services are online in the ePO - On-prem role. Only run the update on one node in the cluster.

- After successful installation of the update on the first node, perform the cluster failover to log on to the second node. Perform the repair operation on the second node to complete the ePO - On-prem Update installation.
- Click Finish.

 Note

For ePO - On-prem in a Windows cluster, make sure that the generic services are online in ePO - On-prem role. Starting the services might take a few minutes to complete.

- (Optional) Repeat these steps on all Agent Handlers.

Install the latest build on your ePO - On-prem server

Follow these steps to install the latest build of ePO 5.10.0 Service Pack1 Update over the older build of ePO 5.10.0 Service Pack1 Update.

- Make sure your system meets these requirements before installing the latest update:

- ePO - On-prem 5.10.0 RTS/EA build is installed.
 - Administrator permissions are needed.
 - Stop the ePO - On-prem Application Server Service (Tomcat).
1. Log on to ePO - On-prem as administrator.
 2. Copy and extract the update package to your ePO - On-prem server.
 3. Run ePOUpdater.exe from the folder where you extracted the update package.
 4. Enter the database password and click Continue.
 5. Select Previously installed Updates and click **Repair**.
 6. Click Finish.

Repair the latest update on your ePO - On-prem server

Follow these steps to latest update on previously applied Service pack 1 Update.

- Make sure your system meets these requirements before installing the latest update:
 - ePO - On-prem 5.10.0 RTS/EA build is installed.
 - Administrator permissions are needed.
 - Stop the ePO - On-prem Application Server Service (Tomcat).
1. Log on to ePO - On-prem as administrator.
 2. Copy and extract the update package to your ePO - On-prem server.
 3. Run ePOUpdater.exe from the folder where you extracted the update package.
 4. Enter the database password and click Continue.
 5. Select Previously installed Updates and click **Repair**.
 6. Click Finish.

Install the latest update on Remote Agent Handlers

Run the Updater tool on the ePO - On-prem server before running it on Remote Agent Handlers.

1. Run the Updater tool on the ePO - On-prem server.

2. Run the Updater tool on Remote Agent Handlers.
3. Make sure that these services aren't running.
 - Agent Handler Service including Remote Agent Handlers (Apache)
 - Event Parser Service
4. Start the Updater tool.

Verify the update installation

Verify that the update successfully completed.

1. From ePO - On-prem, select Menu → Server Settings → Server Information.
2. Verify the latest update number appears in the Update Installed as **Service pack 1**.

Roll back updates

ePO - On-prem updates allow you to roll back an update without going through a disaster recovery process. You can only roll back the last update you applied.

Stop these services:

- ePO - On-prem Application Server Service (Tomcat)
- Agent Handler Service including Remote Agent Handler Service (Apache)
- Event Parser Service

1. Log on to ePO - On-prem as administrator.
2. Copy and extract the downloaded update package to your ePO - On-prem server.
3. Run ePOUpdater.exe from the folder where you extracted the update package.
4. Click Rollback. Verify if the Rollback button changes to Successfully Rolled Back in green.
5. Click Finish.
6. Verify the entry for Update Version doesn't appear on the ePO - On-prem console under Menu → Server Settings → Server Information.
7. Verify the previously applied update number appears in the Update Installed.

 Note

These steps apply to all Remote Agent Handlers; perform these steps on each Agent Handler in your environment after you rollback from the latest update.

Use Silent Mode to install a ePO - On-prem update

Use the command line to install scripted files in Silent Mode on the ePO - On-prem 5.10 server and Agent Handler servers.

 Note

This feature is only supported on Update 5 and later.

1. Log on as administrator to the server hosting ePO - On-prem.
2. Copy and extract the ePO - On-prem update .zip package to the server hosting ePO - On-prem.
3. Run the Command Prompt as administrator:
 - a. Run the install.bat file, located here: `epo_5.10.0_servicepack1update\resources\app`.

For example, `C:\ePO CU BUILDS\epo_5.10.0_servicepack1update\resources\app\install.bat`.
 - b. Enter the command `install.bat [your password]` and replace the default password with your ePO - On-prem database server password.

For example, `C:\Users\cloudadmin.WIN-QFN79SPC5U4.000\Desktop\CU BUILDS\epo_5.10.0_servicepack1update\resources\app> install.bat Welcome123`.
4. Verify the installation:
 - If the latest update is installed, the message Update is already installed appears.
 - If the update isn't installed, the service restarts and the installation runs.

For more information, see [Verify the update installation](#).

Troubleshooting and log file reference

The most common messages that appear while installing ePO - On-prem during an installation and their solutions are listed here. Use this information to troubleshoot problems with your installation.

If you are unable to resolve an issue using the information in this section, contact Technical Support after you have taken these steps:

1. Verify that you have met the minimum installation requirements.
2. Review the release notes and click the link to the Trellix Knowledge Base article to see any known installation issues.
3. Verify that the account you used to log on to the computer where you are installing the software has full administrator permissions to that computer.
4. Collect the exact text of all messages and write down any message codes that appear.
5. Gather the installation log files from C:\Program Data\Trellix\ePO.

Common installation messages with their causes and solutions

ePO - On-prem provides feedback during installation that might require additional action. Review this table for more information about actions required if these messages appear.

Message	Cause	Solution
You are attempting to upgrade from a product version that is not supported.	No version of ePO - On-prem software has been installed on this computer. You can only upgrade from a supported version of ePO - On-prem server.	Select an appropriate installation option.
Internet Explorer 8.0 or later, or Firefox 10 must be installed for this installation to continue.	The computer where you are trying to install the software is using an unsupported version of the browser.	Install a supported Internet browser before continuing.
Another instance of the ePO - On-	The ePO - On-prem setup	Allow the first instance of the

Message	Cause	Solution
prem installer is already running.	program is already running. You can't run more than one instance of the installer at a time.	installer to finish, or stop the first instance and restart your installation.
For security reasons, Trellix does not allow blank passwords. Please type a valid password to continue.	The Password box is blank.	Specify the password of the user account that you want to use.
We recommend that you set the video display resolution to 1366x768 or higher.	The computer where you are trying to install the software does not meet the minimum monitor resolution requirement.	Change the monitor resolution to 1024x768 or higher, then continue the installation. Otherwise, you might not be able to view the whole screen after you start the software. For instructions on changing the monitor resolution, see the Windows Help file (click Start, then select Help).
We recommend that you install the software on a computer with at least 8 GB of RAM.	The computer where you are trying to install the software does not meet the minimum memory requirement.	Add more memory to your system or select a different system for installation that has at least 8 GB of RAM.
ePO - On-prem software requires that your computer is running Windows Server 2008, Windows Server 2012, or Windows Server 2016.	The computer where you are trying to install the software is using a non-supported version of the operating system.	Use a supported server-class operating system.
Enter a value in the "Agent Broadcast communication" field.	The Agent Broadcast communication port box is blank.	Specify the port number (default is 8082) that the ePO - On-prem server uses to send wake-up calls to SuperAgents.
Enter a value in the "Agent-to-	The Agent-to-Server	Specify the port number that the

Message	Cause	Solution
Server communication" field.	communication port box is blank.	agent uses to communicate with the server.
Enter a value in the "Agent Wake-Up communication" port.	The Agent Wake-Up communication port box is blank.	Specify the port number (default is 8081) that the ePO - On-prem server uses to send Trellix Agent wake-up calls.
ePO - On-prem must be installed in a folder. Enter a Destination Folder to continue.	The Destination Folder box is blank or shows the root of a drive.	Click Browse to select a location. The default location is: C:\Program Files\Trellix\ePolicy Orchestrator.
Enter a value in the "User Name" field.	The User name box is blank.	Specify the user name of the account that you want to use.
The License file is missing or corrupt. Contact Technical Support for assistance.	Setup is unable to read the license information required to install the software.	Contact Technical Support.
The operating system or Service Pack you are using is not currently supported.	The computer where you are trying to install the software is using a non-supported version of the operating system.	Use a supported server-class operating system.
The passwords you typed do not match. Type a valid password to continue.	The value you typed in Password and Confirm Password do not match.	Specify the password of the account that you want to use.
The ePO - On-prem license has expired.	Your license to use the software has expired.	Contact your administrator or designated Trellix representative.
This system is not currently configured with a static IP address, which is recommended for the ePO - On-prem server.	The computer where you are trying to install the software does not use a static IP address. We recommend using static IP addresses for ePO - On-prem	Specify a static IP address for use with your ePO - On-prem server.

Message	Cause	Solution
	servers to improve performance and reduce bandwidth use.	
Unable to make a connection to the database server. Verify that you provided the account credentials and database server name correctly, then try again.	A connection could not be made to the corresponding ePO - On-prem database server.	1. Verify that the Domain, User Name, and Password you provided are typed correctly. 2. Verify that the database server is running. 3. Verify that the user account you provided is valid for the database server.
Unable to connect using the information you provided. Verify that you entered the correct information and try again.	The user account that you specified could not be accessed.	1. Verify that the Domain, User Name, and Password you provided are typed correctly. 2. Verify that the account you used to log on to this computer has access to this domain.

Log files for troubleshooting

ePO - On-prem provides log files that contain important information when troubleshooting.

These log files are separated into three categories:

- **Installer logs** — Include details about installation path, user credentials, database used, and communication ports configured.
- **Server logs** — Include details about server functionality, client event history, and administrator services.

- **Agent logs** — Include details about agent installation, wake-up calls, updating, and policy enforcement.

Installer logs

Installer log files list details about the ePO - On-prem installation process.

These logs provide information about:

- Actions taken by specific components
- Administrator services used by the server
- Success and failure of critical processes

File name	Log type	Location	Description
AH5100-Install-MSI.log	Agent Handler installation	C:\ProgramData\McAfee\ ePolicy Orchestrator	This file logs all Agent Handler installation details including: • Installer actions • Installation failures
AH5100-ahetupdll.log	Temporary	%temp% (on the Agent Handler server)	Logs Agent Handler back-end events.
core-install.log	Temporary	%temp%\TrellixLogs\ eP05100-Troubleshoot\MFS	Generated when the installer calls the MFS ANT installer. Provides information about: • Creation of server database tables • Installation of server components  Note: This file is deleted if the installation succeeds.

File name	Log type	Location	Description
epo-install.log	Installation	C:\ProgramData\McAfee\epolicy Orchestrator	Created when the installer calls the ANT installer.
EP05100-Checkin-Failure.log	Installation	C:\ProgramData\McAfee\epolicy Orchestrator	Generated when installer fails to check in any of these package types: • Extensions • Plug-ins • Deployment packages • Agent packages
EP05100-CommonSetup.log	Installation	C:\ProgramData\McAfee\epolicy Orchestrator	Contains installer details such as: • Custom Action logging • SQL, DTS (Microsoft Data Transformation Services), and service-related calls • Registering and unregistering DLLs • Files and folders selected for deletion at restart
EP05100-Install-MSI.log	Installation	C:\ProgramData\McAfee\epolicy Orchestrator	The primary installation log. Contains installation details such as installer actions and installation failures.

File name	Log type	Location	Description
<ExtensionFileName>.cmd	Temporary	%temp%\TrellixLogs\ eP05100-troubleshoot\ OutputFiles	Created by the installer. Contains the command (sent to Remote-Client) to check in extensions.  Note: If the installation succeeds, these files are deleted.
MFS5100-CommonSetup.log	Installation	C:\ProgramData\McAfee\ ePolicy Orchestrator	Contains core functionality installer details.

Server logs

Server log files contain details about server functionality and various administrator services used by ePO - On-prem.

File name	Log type	Location	Description
EpoApSvr_<serverName> .log	Primary	[InstallDir]\DB\Logs	Application Server log file with details about repository actions such as: • Pull tasks • Checking in deployment packages to the repository • Deleting deployment packages from the repository  Note: This file is not present until after initial service startup.
Errorlog.<CURRENT_DATE TIME>	Apache	[InstallDir]\Apache2\ logs	Contains Apache service details.

File name	Log type	Location	Description
			 Note: This file is not present until after the Apache service is started for the first time.
Eventparser_<serverName>.log	Primary	[InstallDir]\DB\Logs	Contains ePO - On-prem event parser services details, such as product event parsing success or failure.
Jakarta_service_<DATE>_<serverName>.log	Tomcat	[InstallDir]\Server\logs ^[a]	Contains ePO - On-prem Application Server service details.  Note: This file is not present until after the initial Tomcat service startup.
Localhost_access_log.<DATE>.txt	Tomcat	[InstallDir]\Server\logs [*]	Records all ePO - On-prem server requests received from client systems.  Note: This file is not present until after the initial Tomcat service startup.
Orion_<serverName>.log	Primary	[InstallDir]\Server\logs [*]	Contains platform details and all extensions loaded by default.  Note: This file is not present until after the ePO - On-prem Application Server service is started for the first time.

File name	Log type	Location	Description
Replication_<serverName>.log	Server	[InstallDir]\DB\Logs	The ePO - On-prem server replication log file. This file is generated only when all these criteria are true: • There are distributed repositories. • A replication task has been configured. • A replication task has run.
Server_<serverName>.log	Primary	[InstallDir]\DB\Logs	Contains details related to these ePO - On-prem server services: • Agent-server communications • ePO - On-prem Server Agent Handler  Note: This file is not present until after initial service startup.
Stderr_<serverName>.log	Tomcat	[InstallDir]\Server\logs*	Contains any Standard Error output captured by the Tomcat service.  Note: This file is not present until after the initial Tomcat service startup.
<AgentGuid>_<Timestamp>_Server_manifest.xml	Policy	[InstallDir]\DB\DEBUG	Contains details about policy updating issues. To enable this file: 1. Browse to this registry key: HKEY_LOCAL_MACHINE\Software\NetworkAssociates\ePolicyOrchestrator\ 2. Create this DWORD with value

File name	Log type	Location	Description
			1: SaveAgentPolicy 3. Restart the Trellix ePolicy Orchestrator - On-premises Server (Apache) service.  Tip Enable this file for the minimum time to capture the required information, because the resulting files grow rapidly.
[a] In cluster environments, the log file is at [InstallDir]\Bin\Server\logs.			

Trellix Agent logs

Trellix Agent log files contain actions triggered or taken by the Trellix Agent.

File names in this list reflect Trellix Agent version 5.5.0 for Windows.

Trellix Agent 5.5.0 file name	Log type	Location	Description
masvc_<hostname>.log	Server	[Agent DATA Path]\logs	Generated when masvc.exe is used. The file contains information related to: - Property collection - Policy enforcement - Scheduling of tasks - Agent server

Trellix Agent 5.5.0 file name	Log type	Location	Description
			communication Update sessions
macmnsvc_<hostname>.log	Trellix Agent	[Agent DATA Path]\logs	Generated when macmnsvc.exe is used. The file contains information related to: - Peer-to-Peer server - SuperAgent - Wake-up - RelayServer
macompatsvc_<hostname>.log	Trellix Agent	[Agent DATA Path]\logs	Generated when macompatsvc.exe is used. The file contains information related to the compatibility of managed products with Trellix Agent services.
masvc_<hostname>_backup_<backupcountnumber>.log	Trellix Agent	[Agent DATA Path]\logs	Generated as backup files for agent services.
marepomirror.log	Server		Generated when marepomirror.exe is used. The file contains information related to mirroring of the repository.
FrmInst_<hostname>.log	Trellix Agent	%temp%\McAfeeLogs	Generated when FrmInst.exe is used to install the Trellix Agent. This file contains: Informational

Trellix Agent 5.5.0 file name	Log type	Location	Description
			messages - Progress messages - Failure messages if installation fails
McScript.log	Trellix Agent Debug	[Agent DATA Path]\logs	Contains the results of script commands used during agent deployment and updating. To enable the DEBUG mode for this log, set this DWORD value on the client's registry key: HKEY_LOCAL_MACHINE\SOFTWARE\NETWORK ASSOCIATES\TVD\SHARED COMPONENTS\FRAMEWORK\DWDEBUGSCRIPT=2  Note: Delete this key when you finish troubleshooting.
MFEAgent.msi. <system time stamp>.log	Trellix Agent	%temp%\McAfeeLogs	Contains details about the MSI installation of the agent.
UpdaterUI_<system>.log	Trellix Agent	%temp%\McAfeeLogs	Contains details about the updates to managed products on the client system.

Trellix Agent error logs

When the Trellix Agent traps errors, they are reported in Trellix Agent error logs. These error logs are created at %temp%\McAfeeLogs during installation. Trellix Agent error logs are named using the convention <filename>_error.log.

For example, when performing client tasks, information is normally logged in McScript.log. Errors that occur are logged in McScript_error.log.

Error logs only contain details about errors.

Note

After installation, the Trellix Agent logs are located in %ProgramData%.

Access your ePO - On-prem with your ePO - SaaS account

If you have an active ePO - SaaS account, you can access your ePO - On-prem through your ePO - SaaS account. To access, you need to first connect your SaaS and On-prem accounts.

If you are using Cloud Bridge version...	Use this method
2.2 (also if you are a Federal customer)	ePO - SaaS registration token You can link your ePO - On-prem by logging into your ePO - SaaS account and generating a registration token of client type ePO - On-prem.
2.1 or 2.0	ePO - SaaS credentials authentication You can link your ePO - On-prem by adding your ePO - SaaS account credentials in your ePO - On-prem account.

Download and install ePO - SaaS Cloud Bridge

You can download the ePO - SaaS Cloud Bridge extension from the Trellix Software Download site using your grant number.

Download the cloud bridge extension

You must have a valid grant number to access and download the Trellix products.

1. Using your browser, access the Trellix [Product Downloads](#) site.
2. In Product Downloads, click Download to open the Download My Products Login page.
3. Type your grant number and the displayed CAPTCHA code, then click Submit.
4. Search for Trellix ePolicy Orchestrator - On-premises and select version 5.10.0.
5. In Available Downloads, Select Filters type as EXTENSION and select ePO - SaaS Cloud Bridge, and download it to your local ePO - On-prem server.

 Note

You can also download the extension from the Software Manager of your ePO - On-prem server. For more information about downloading extensions from the Software Manager, see the product guide of your ePO - On-prem version.

Install the extension

Install the ePO - SaaS Cloud Bridge extension that connects your ePO - On-prem server to your ePO - SaaS account.

Make sure that you downloaded the **CloudBridge2_release_<version_number>.<build_number>.zip** extension file.

1. Log on to the ePO - On-prem server as an administrator.
2. Select Menu → Software → Extensions → Install Extension.
3. Browse to and select the CloudBridge2_release_<version_number>.<build_number>.zip extension file, then click OK.
4. Verify that the ePO - SaaS Cloud Bridge name appears in the Extensions list.

 Note

You can also download the extension from the Software Manager of your ePO - On-prem server and follow these instructions to install the extension.

Now you can configure the ePO - SaaS Cloud Bridge server settings.

Configuring ePO - SaaS Cloud Bridge using Registration token

When you configure the ePO - SaaS Cloud Bridge server settings, you can link your ePO - On-prem server to your ePO - SaaS account.

Using ePO - SaaS registration token

The ePO - SaaS Cloud Bridge extension must be installed on your ePO - On-prem server.

 Note

You must have a valid ePO - SaaS account with **Trellix Account Administrator** role assigned.

1. From ePO - SaaS, select Menu → Configuration → Appliance and Server Registration.
 - a. Click Add to add a new registration token.
 - b. Select the Client type as Trellix ePolicy Orchestrator - SaaS.

- c. Type the number of appliances or servers you want to register in the Number of clients field and click **Save**.

To view the generated token, expand Trellix ePolicy Orchestrator - SaaS under **Servers**. Note down the registration token.

- d. Perform one of these actions if you want to revoke the token:
 - From the token list, click **Revoke** under Actions.
 - Select the token, then select Revoke from the Take Action drop-down list in the token details pane.
 - To revoke the token from a client - select a token from the token list, then select one or more clients from the Registered Clients pane and select **Revoke** from the Take Action drop-down list.

2. Log on to the ePO - On-prem server as an administrator. Select Menu → Configuration → Server Settings, then select Trellix ePO-SaaS Cloud Bridge from Setting Categories. The Trellix ePO-SaaS Cloud Bridge Server Settings page displays these options:

- a. Status— Displays the status of the connection. See *Status Messages* for detailed status information.



Note

Before you configure your connection, the status is **This server is not linked with Trellix ePO-SaaS**.

- b. Tenant ID — Displays the tenant ID of the linked ePO-SaaS account.
- c. Cloud Bridge Client ID — Displays the Client ID associated with the token.

3. From the Trellix ePO-SaaS Cloud Bridge Server Settings page, click Edit.

- a. Enter the Registration Token generated from Step 1.
- b. In the **ePO Logon URL** section, enter the ePO - On-prem URL which you are using to access the on-premises ePO - On-prem server.
- c. Click **Save**.



Note

The Cloud Bridge settings page now contains a field ePO Logon URL which can be seen only from ePO - On-prem 5.10.0 Update 7 onwards. This URL is added to the allowed URL list in Identity and Access Management (IAM). This is needed for IAM to redirect the user to the on-premises ePO - On-prem after authentication using [Log On With ePO - SaaS](#) feature.

4. The ePO - SaaS Cloud Bridge Server Settings page displays the status as This server is actively linked, and the linked account as the email address for the Trellix account:

- a. Status — After configuration, above the Refresh button, the status is **This server is actively linked**.
- b. Trellix ePO-SaaS Tenant ID — The ID for the particular ePO - SaaS tenant.
- c. **Cloud Bridge Client ID** — The Client ID associated with the token.
- d. **ePO Logon URL** — Your ePO - On-prem URL which you are using to access the ePO - On-prem server.

Your ePO - On-prem server is now connected to the ePO - SaaS account.

Note

You might see some error messages while linking your ePO - On-prem account with the ePO - SaaS account. For more information, see [Common error messages while linking ePO - On-prem with ePO - SaaS account](#).

5. To view the connected servers in ePO - SaaS, go to Menu → Configuration → Appliance and Server Registration.
 - a. Click Trellix ePolicy Orchestrator - SaaS under Servers.
 - b. The Used Clients column displays the number of clients connected to ePO - SaaS.

Download and install ePO - SaaS Cloud Bridge

You can download the ePO - SaaS Cloud Bridge extension from the Trellix Software Download site using your grant number.

Download the cloud bridge extension

You must have a valid grant number to access and download the Trellix products.

1. Using your browser, access the Trellix [Product Downloads](#) site.
2. In Product Downloads, click Download to open the Download My Products Login page.
3. Type your grant number and the displayed CAPTCHA code, then click Submit.
4. Search for Trellix ePolicy Orchestrator - On-premises and select version 5.10.0.
5. In Available Downloads, Select Filters type as EXTENSION and select ePO - SaaS Cloud Bridge, and download it to your local ePO - On-prem server.

Note

You can also download the extension from the Software Manager of your ePO - On-prem server. For more information about downloading extensions from the Software Manager, see the product guide of your ePO - On-prem version.

Install the extension

Install the ePO - SaaS Cloud Bridge extension that connects your ePO - On-prem server to your ePO - SaaS account.

Make sure that you downloaded the **CloudBridge2_release_<version_number>.<build_number>.zip** extension file.

1. Log on to the ePO - On-prem server as an administrator.
2. Select Menu → Software → Extensions → Install Extension.
3. Browse to and select the CloudBridge2_release_<version_number>.<build_number>.zip extension file, then click OK.
4. Verify that the ePO - SaaS Cloud Bridge name appears in the Extensions list.



You can also download the extension from the Software Manager of your ePO - On-prem server and follow these instructions to install the extension.

Now you can configure the ePO - SaaS Cloud Bridge server settings.

Configure using ePO - SaaS credentials authentication

Once you have downloaded and installed the ePO - SaaS Cloud Bridge extension on your ePO - On-prem server, you can then configure your ePO - On-prem to enable log on using ePO - SaaS credentials.

- Ensure that you have a valid ePO - SaaS account with **Trellix Cloud Administrator** permissions.
- If you have other Identify Provider configured in ePO - SaaS, note that you cannot use the IdP credentials to configure ePO - On-prem logon. You must use a valid customer ID.

1. Log on to the ePO - On-prem server as an administrator.
2. Select Menu → Configuration → Server Settings, then select Trellix ePO - SaaS Cloud Bridge from Setting Categories. The Trellix ePO - SaaS Cloud Bridge Server Settings page displays these options:

- Status — Displays the status of your ePO - SaaSConnection. See *Status Messages* for detailed status information.



Before you configure your connection, the status is **Not Linked**.

- Refresh — Refreshes the interface with the current data and status from the ePO - On-prem database.
- Linked Account — Displays the email address of the linked ePO - SaaS account.

3. Add the ePO - SaaS users who want to access ePO - On-prem via [Log On With ePO - SaaS](#) using the following steps —

- Select **Menu** → **User Management** → **User**, then click **New User** button.
- In the User name field, type the email address of your ePO - SaaS account.
- Under **Authentication type**, select **ePO - SaaS Authentication**.
- Under **Manually assigned permission sets**, assign a permission set for the user.
- Click **Save**.

Upon successful creation of the user, log off from the ePO - On-prem console.

4. From the Trellix ePO - SaaS Cloud Bridge Server Settings page, click Edit.

- Enter the ePO - SaaS account email and password.
- In the **ePO Logon URL** section, enter the ePO - On-prem URL which you are using to access the on-premises ePO - On-prem server.
- Click **Save**.

Note

The ePO - SaaS Cloud Bridge settings page now contains a field ePO Logon URL which can be seen only from ePO - On-prem 5.10.0 Update 7 onwards. This URL is added to the allowed URL list in Identity and Access Management (IAM). This is needed for IAM to redirect the user to the on-premises ePO - On-prem after authentication using [Log On With ePO - SaaS](#) feature.

5. The ePO - SaaS Cloud Bridge Server Settings page displays the status as This server is actively linked, and the linked account as the email address for the SaaS account:

- Status — After configuration, the status is modified to **This server is actively linked**.
- Linked Account — The email address configured for the ePO - SaaS account.
- **Trellix ePO - SaaS Customer ID** — The ePO - SaaS Customer ID that is linked.
- **ePO Logon URL** — Your ePO - On-prem URL which you are using to access the ePO - On-prem server.

Your ePO - On-prem server is now connected to the ePO - SaaS account.

Note

You might see some error messages while linking your ePO - On-prem account with the ePO - SaaS account. For more information, see [Common error messages while linking ePO - On-prem with ePO - SaaS account](#).

Error messages

You might see some error messages during the ePO - SaaS Cloud Bridge linking process.

Not all error messages appear in the product interface. But, you can see the related messages in `orion.log`.

This table lists error messages and their description for troubleshooting.

Error messages	Description	Suggested solutions
Error occurred while getting Client ID and Secret from IAM service.	1. The Trellix service might have failed to process the provided account data. 2. If ePO server is behind the proxy network and Proxy is not added in ePO server. 3. If URL's related to ePO - SaaS Cloud Bridge are blocked in the customer's firewall.	1. Contact Trellix Support. 2. Add proxy settings in ePO - On-prem server settings. see KB94929. 3. Allow ePO - SaaS Cloud Bridge related URL's in customer network firewall. See KB94930.
Registration token is not valid.	This error occurs if you enter Invalid registration token while linking the Cloud Bridge, or if the token has expired.	Confirm the registration token entered is correct.
Registration token has either already been used for Max Clients or has been revoked. Try again with a new registration token.	1. You enter registration token which has been revoked from ePO - SaaS Appliance and Server Registration. 2. Registration token is used in more clients	You need to generate new registration token and use it for Cloud Bridge account linking.

Error messages	Description	Suggested solutions
	than it is intended for (Max Client count in Appliance and Server Registration).	
Cloud Bridge status gets updated to 'This server is linked but the Cloud Bridge Client ID is revoked. Relink to Trellix'.	Cloud Bridge status gets updated because of revoking the Cloud Bridge client from ePO - SaaS. After revoking the Cloud Bridge client, ePO - On-prem console shows the notification with 'ePO - SaaS Cloud Bridge client has been revoked' message.	Relink your ePO server to your Trellix account to ensure uninterrupted communication with Trellix.
Existing linked tenant could not be unlinked.	When you try to unlink the already linked ePO - On-prem in Cloud Bridge setting, you see this error if that unlink process failed for some reason.	Check the orion.log file for information about a particular error message or contact Trellix Support .
An error occurred while linking the tenant.		Check the orion.log file for information about a particular error message or contact Trellix Support .

Adding an SSL certificate to trusted collection

Supported browsers warn about a server's SSL certificate if it cannot verify the certificate.

By default, the ePO - On-prem server uses a self-signed certificate for SSL communication with the browser, which the browser does not trust. A warning message displays every time you visit the ePO - On-prem console.

To stop this warning message from appearing, do one of the following:

- Add the ePO - On-prem server certificate to the browser's collection of trusted certificates.
- Add the certificate for every browser that interacts with ePO - On-prem. If the browser certificate changes, add the server certificate again.
- (Recommended) Replace the default ePO - On-prem server certificate with a valid certificate signed by a certificate authority (CA) that the browser trusts. You only need to add the certificate once for web browsers in your environment.
- If the server host name changes, replace the server certificate with a new trusted CA certificate.

For more information, see [KB72477](#).

To replace the ePO - On-prem server certificate, you must first obtain the certificate signed by a trusted CA. You must also obtain the certificate's private key and its password (if it has one). Then you can use all these files to replace the server's certificate.

The ePO - On-prem server expects the server certificate to use these formats: PKCS7, PEM encoded, DER encoded, or PKCS12 file with extensions .cer, .crt, .p12, or .p7b.

The ePO - On-prem browser expects the linked files to use PEM for private keys.

If the server certificate or private key is not in these formats, convert to one of the supported formats before replacing the default certificate.

If your organization requires a higher standard of encryption, replace the default SHA-256 certificate with one that uses SHA-384 or higher.

Replace the server certificate

Update the default server certificate used for HTTPS communication with browsers.

You must have access to the new certificate and private key files.

1. Open the Edit Server Certificate page.
 - a. Select Menu → Configuration → Server Settings.
 - b. From the Setting Categories list, select Server Certificate, then click Edit.

2. Browse to the server certificate file, then click Open.

 Note

You can create your own self-signed certificate with OpenSSL.

3. If needed, type the PKCS12 certificate password and alias name.
4. Browse to the private key file, then click Open.
5. If needed, type the private key password, then click Save.
6. Restart ePO - On-prem for the change to take effect.

Install the security certificate for Internet Explorer

Prevent the certificate prompt from appearing every time you log on by installing the security certificate.

1. From your browser, open ePO - On-prem. The Certificate Error: Navigation Blocked page appears.
2. Click Continue to this website (not recommended) to open the logon page. The address bar is red, indicating the browser cannot verify the security certificate.
3. To the right of the address bar, click Certificate Error to display the Certificate Invalid warning.
4. At the bottom of the warning, click View certificates to open the Certificate dialog box.

 Caution

Do not click Install Certificate on the General tab. If you do, the process fails.

5. Select the Certification Path tab, then select Orion_CA_<servername>, and click View Certificate. Another dialog box opens to the General tab, displaying the certificate information.
6. Click Install certificate to open the Certificate Import Wizard.
 - a. Click Next to specify where the certificate is stored.
 - b. Select Place all certificates in the following store, then click Browse to select a location.
 - c. Select the Trusted Root Certificate Authorities folder from the list, click OK, then click Next.
 - d. Click Finish. In the Security Warning that appears, click Yes.
7. Close the browser.

8. Change the target of the ePO - On-prem desktop shortcut to use the NetBIOS name of the ePO - On-prem server instead of "localhost."
9. Restart ePO - On-prem.

Now when you log on to ePO - On-prem, you are no longer prompted to accept the certificate.

Install the security certificate for Firefox

You can install the security certificate when using Firefox 3.5 or later, so that the warning dialog box does not appear every time you log on.

1. From your browser, open ePO - On-prem. The This Connection is Untrusted page appears.
2. Click I Understand the Risks at the bottom of the page.
3. Click Add Exception.
4. Click Get Certificate. The Certification Status information is populated and the Confirm Security Exception button is enabled.
5. Make sure that Permanently store this exception is selected, then click Confirm Security Exception.

Now when you log on to ePO - On-prem, you are no longer prompted to accept the certificate.

Install Agent Handlers

Install Agent Handlers in your environment to help manage agent-server communication and load balancing. You can install Agent Handlers at any time.

Update the system with the latest Microsoft security updates, then turn off Windows updates during the installation process.

Each ePO - On-prem server contains a main Agent Handler. Installing more Agent Handlers can help manage an increased number of products and systems managed by one ePO - On-prem server in situations where the CPU and IO on the database server is not overloaded.

Agent Handlers require the same high-speed network access to your database as the primary ePO - On-prem server.

Important

Install additional Agent Handlers in the same data center as the SQL Server. Do not install Agent Handlers in remote locations or you risk impacting the performance of the entire ePO - On-prem environment.

To use more IP addresses for agent-server communication, create an Agent Handler group, and add the additional IP address to the virtual IP address input field.

1. Open the folder where you extracted the contents of the ePO - On-prem software installation package.
2. Copy the Agent Handler folder to the intended Agent Handler server system.
3. Right-click Setup.exe and select Run as Administrator to start the Trellix Agent Handler InstallShield wizard.

After some installation activities take place in the background, the InstallShield wizard opens. Click Next to begin the installation process.

4. Accept the terms in the license agreement.

The Destination Folder step opens

5. Accept the default destination or click Change to select a different destination, then click Next.

Important

The destination path must not contain double-byte characters. The path characters are a limitation of the Apache web server. Using double-byte characters causes the installation to fail and the Apache web server service to fail on startup.

6. Configure the server information.

- a. Type the system name of the ePO - On-prem server with which the Agent Handler must communicate.
- b. Specify which port to use for Agent Handler-to-server communication. The default port is 8443, the same port used for client-to-server authenticated communication.
- c. Type the name and password of a user with ePO - On-prem Global Administrator rights, and click Next.
- d. Provide the password for access to the ePO - On-prem SQL database, then click Next.

The Database Information page is populated with these ePO - On-prem server settings.

- Database Server with instance name. For example, **DB-SERVER\SERVERNAME**.
- Authentication type.
- Domain name where the database server is hosted.
- User name and Password.
- Database name if not provided automatically.

7. Click Install to start the installation.

8. When installation is complete, enable your Agent Handler from the ePO - On-prem interface.

Disaster Recovery Snapshot prerequisites

Make sure these requirements are in place before you begin restoring ePO - On-prem from a Disaster Recovery Snapshot.

- ePO - On-prem SQL database containing a valid Disaster Recovery Snapshot.
- User name and password for a global administrator account that uses ePO - On-prem authentication.
- The Disaster Recovery passphrase for the snapshot in the database.
- Install ePO - On-prem 5.10.2428.67 or later in Primary Node after enabling the Restore DB option.
- Failover to the secondary node and complete the installation without enabling the Restore DB option. During this process expect unavailability of Tomcat service.
- If you apply ePO - On-prem 5.10.0 Update 13 after taking snapshot on the previous ePO - On-prem 5.10.0 server, you must perform disaster recovery with ePO - On-prem 5.10.2428.67 or later, then perform the repair function with ePO - On-prem 5.10.0 Update 13.
- When snapshot is taken on ePO - On-prem 5.10.0 Update 13, during Disaster Recovery restore, ePO package 5.10.0.2428.67 or later automatically performs the repair for Update 13, that means manual repair is not needed post Disaster Recovery restore.

For more details about the Disaster Recovery process, see the product guide for ePO - On-prem.

Restore ePO - On-prem software in a single-server environment

Restore ePO - On-prem from the snapshot stored in a ePO - On-prem database. You can do this by reinstalling the ePO - On-prem software on a server with the Restore ePO from an existing database option enabled and configuring the installation to use an existing ePO - On-prem database.

Gather this information and complete these steps before beginning your installation. These steps make sure that your ePO - On-prem software can communicate with the SQL Server hosting the ePO - On-prem database.

- If you are using dynamic ports for your SQL Server, verify that the SQL Browser Service is running.
- If you are not using dynamic ports for your SQL Server, make sure that you know the ports that your SQL instance is using.
- Make sure that the TCP/IP Protocol is enabled in the SQL Server Configuration Manager.

 Note

Trellix Agent uses either the last known IP address, DNS name, or NetBIOS name of the ePO - On-prem server. If you change any one of these values, make sure that Trellix Agent has a way to locate the server. The easiest way to do that is to retain the existing DNS record, and change it later to direct the ePO - On-prem server's new IP address. After Trellix Agent successfully connects to the ePO - On-prem server, it downloads an updates `SiteList.xml` with the current information.

 Note

We recommend that you change the FQDN first, and use the same IP address so that Trellix Agent on the end nodes communicates to the Agent Handler or the ePO - On-prem server using the last known IP address. After successful communication, Trellix Agent will update the new FQDN of the server. Once all systems communicate successfully using the IP address, you can change the IP address as Trellix Agent knows the new FQDN.

 Note

Monitor the process because you might need to restart your system.

1. If you have Agent Handlers configured, log on to the systems where the Agent Handlers are installed, then open the Windows Services panel and stop the Trellix Event Parser and Trellix Apache services.

See your Microsoft software product documentation for more information about using the Windows Services panel.
2. Using an account with local administrator permissions, log on to the Windows Server that you want to restore ePO - On-prem to.
3. Extract the files to a temporary location, and double-click Setup.exe.

The version of ePO - On-prem being restored must be the same as the version used to create the snapshot in the database. You can download the correct version from the Trellix website.

 Important

If you try to run Setup.exe without first extracting the contents of the .zip file, the installation fails.

The Trellix ePolicy Orchestrator-On-prem - InstallShield Wizard starts.

4. Select Restore ePO from an existing database snapshot and click Next to begin the installation process.
5. In the Install additional software step, any remaining prerequisites are listed. To install them, click Next.
6. In the Destination Folder step, click either:
 - Next — Install your ePO - On-prem software in the default location (C:\Program Files (x86)\Trellix\ePolicy Orchestrator - On-premises).

- Change — Specify a custom destination location for your ePO - On-prem software. When the Change Current Destination Folder window opens, browse to the destination and create folders as needed. When finished, click OK → Next.
7. In the Database Information step, either select the SQL Server name from the Database Server drop-down list or manually enter the name of the SQL Server.
 8. In the Database Name field, enter the name of the existing ePO - On-prem database containing the snapshot, specify the type of Database Server Credentials to use, then click Next.
 - Windows authentication — From the Domain menu, enter the domain of the user account you're going to use to access the SQL Server. Enter the user name and password for an account with sufficient permissions to access the SQL Server hosting the ePO - On-prem database.
 - SQL authentication — Enter the user name and password for an account with sufficient permissions to access the SQL Server hosting the ePO - On-prem database.

The Domain menu is grayed out when using SQL authentication.

You might need to specify the SQL Server TCP port to use for communication between your ePO - On-prem server and database server. The ePO - On-prem installation tries to connect using the default TCP port 1433, and to determine if a dynamic port is in use by querying the SQL Browser service on UDP port 1434. If those ports fail, you are prompted to provide a SQL Server TCP port.

9. In the HTTP Port Information step, review the default port assignments, then click Next to verify that the ports are not already in use on this system.
10. In the Administrator Information step, type the user name and password you used for your previously existing ePO - On-prem global administrator account.
11. Type the Keystore Encryption passphrase (also known as the Snapshot passphrase) for the snapshot in the ePO - On-prem database.
12. Click Install to begin the installation.
13. When installation is complete, click Finish to exist the InstallShield wizard.
14. If you restored ePO - On-prem to a server with a different IP address and DNS name than your previously existing server, configure a way to allow your managed systems to connect to your new ePO - On-prem server.

There are several ways to achieve this depending on your DNS vendor. The most common way is to create a CNAME record in DNS that redirects requests to the old DNS name to the IP address of the new ePO - On-prem server. For more information about this process, see Microsoft documentation.

15. If you stopped the Agent Handlers in step 1, and restored ePO - On-prem to a system with the same server name and IP address that it had previously, log on to the systems where the Agent Handlers are installed, then open the Windows Services panel and start the Trellix Event Parser and Trellix Apache services.

If you restored ePO - On-prem to a system with a different name or IP address, see *Restore Agent Handler connections*.

Your ePO - On-prem software is now restored. If needed, double-click the Launch Trellix ePolicy Orchestrator-On-prem icon on your desktop to start using your ePO - On-prem server, or browse to the server from a remote web console (https://<server_name>:<port>).

Restore ePO - On-prem software in a cluster environment

To restore the ePO - On-prem servers installed on server clusters with Microsoft Cluster Server (MSCS) software, reinstall the ePO - On-prem software on all servers in the server cluster.

Gather this information and complete these steps before beginning your installation. These steps ensure that your ePO - On-prem software can communicate with the SQL Server hosting the ePO - On-prem database.

- If you are using dynamic ports for your SQL Server, verify that the SQL Browser Service is running.
- If you are not using dynamic ports for your SQL Server, ensure that you know the ports that your SQL instance is using.
- Make sure that the TCP/IP Protocol is enabled in the SQL Server Configuration Manager.

Restoring the ePO - On-prem software in a Microsoft Cluster Server environment is similar to installing the software initially.



Monitor the Restore installation process. You might need to restart your system.

1. If you have Agent Handlers configured, log on to the systems where the Agent Handlers are installed, then open the Windows Services panel and stop the Trellix Event Parser and Trellix Apache services.

See your Microsoft software product documentation for more information about using the Windows Services panel.

2. Using an account with local administrator permissions, log on to the Windows Server (the first node of the cluster) used as the restore ePO - On-prem server.
3. If needed, in the Failover Cluster Manager, create the ePO - On-prem application role, Client Access Point, and shared data drive resources.

For more information, see *Installing Software in a Cluster environment*.

4. Extract the files to a temporary location, and double-click Setup.exe.

The version of ePO - On-prem being restored must be the same as the version used to create the snapshot in the database. You can download the correct version from the Trellix website.

 Important

If you try to run Setup.exe without first extracting the contents of the .zip file, the installation fails.

The Trellix ePolicy Orchestrator - On-prem - InstallShield Wizard starts.

5. Select Restore ePO from an existing database snapshot and click Next to begin the installation process.
6. In the Install additional software step, any remaining prerequisites are listed. To install them, click Next.
7. In the Destination Folder step, click Change to specify a custom destination location for your ePO - On-prem software. When the Change Current Destination Folder window opens, browse to the destination and create folders as needed. When finished, click OK.

 Important

Make sure that you specify a destination folder that is accessible from all nodes of the cluster.

8. In the Set Virtual Server Settings step, provide the Virtual Server IP address, ePO - On-prem Virtual Cluster Name, ePO - On-prem Virtual Cluster FQDN, and Cluster Configuration Passphrase, then click Next.
9. In the Database Information step, either select the SQL Server name from the Database Server drop-down list or manually enter the name of the SQL Server.
10. In the Database Name field, enter the name of the existing ePO - On-prem database containing the snapshot, specify the type of Database Server Credentials to use, then click Next.
 - Windows authentication — From the Domain menu, enter the domain of the user account you're going to use to access the SQL Server. Enter the user name and password for an account with sufficient permissions to access the SQL Server hosting the ePO - On-prem database.
 - SQL authentication — Enter the user name and password for an account with sufficient permissions to access the SQL Server hosting the ePO - On-prem database.

The Domain menu is grayed out when using SQL authentication.

You might need to specify the SQL Server TCP port to use for communication between your ePO - On-prem server and database server. The ePO - On-prem installation tries to connect using the default TCP port 1433, and to determine if a dynamic port is in use by querying the SQL Browser service on UDP port 1434. If those ports fail, you are prompted to provide a SQL Server TCP port.

11. In the HTTP Port Information step, review the default port assignments, then click Next to verify that the ports are not already in use on this system.
12. In the Administrator Information step, type the user name and password you used for your existing ePO - On-prem global

administrator account.

13. Type the Keystore Encryption passphrase (also known as the Snapshot passphrase) for the snapshot in the ePO - On-prem database.
14. Click Install to begin the installation.
15. When the installation has completed on the first node, click Finish, and move the ePO Application role to the second node.

 Note

You can also shut down the first node, forcing the role to move to the second node.

16. On the second node, run the ePO - On-prem installer.

 Important

Do not select the Restore ePO from an existing database snapshot option.

17. In the Destination Folder step, click Change, browse to the destination on the shared data drive where ePO - On-prem is installed, and click OK → Next.
18. In the Set Virtual Server Settings step, the Virtual Server IP address, ePO - On-prem Virtual Cluster Name, and ePO - On-prem Virtual Cluster FQDN fields are automatically populated. Type the Cluster Configuration Passphrase and click Next.
19. Click Install to start the installation on the second node, then click Finish.

This process takes much less time than the installation on the first node.

20. Create the three Generic Service resources. When finished, bring the ePO Application Role online.

For more information, see *Installing software in a cluster environment*.

21. If you stopped the Agent Handlers in step 1, log on to the systems where the Agent Handlers are installed, then open the Windows Services panel and start the Trellix Event Parser and Trellix Apache services.
22. Confirm that ePO - On-prem is functioning correctly, and test the cluster function by moving the ePO Application Role to the other node.

After completing these steps, your ePO - On-prem software is restored.

Restore Agent Handler connections

If you restored ePO - On-prem to a system with a new name or IP address, you must change the Agent Handler settings to

connect to the restored server.

1. On the Agent Handler server, extract the ePO - On-prem software installation package to a temporary location.
2. In the extracted folder, open the Agent Handler folder, and double-click Setup.exe to start the Trellix Agent Handler InstallShield wizard.
3. Click Next to begin the change process.
4. From the Program Maintenance dialog box, select Modify to change which program features are installed, then click Next.
5. Configure these settings:
 - a. Type the restored system name of the ePO - On-prem server with which the Agent Handler must communicate.
 - b. Specify which port to use for Agent Handler-to-server communication. The default port is 8443.
 - c. Type the ePO Admin User Name and ePO Admin Password of a user with global administrator rights.
 - d. Click Next.

The installer contacts the ePO - On-prem server and obtains the details for the SQL Server hosting the ePO - On-prem database. The SQL Server and database details, except for the password, are already populated.

6. Enter the password for the account you specified and click Next.
7. Click Install to start the changes to the installation.
8. When installation is complete, click Finish.

Your Agent Handlers can now communicate with the restored ePO - On-prem server and SQL database.

FIPS basics

ePO - On-prem provides an operating mode with a higher level of security for environments that require it. This mode (FIPS mode) follows security guidelines detailed in section 140 of the Federal Information Processing Standard (FIPS).

The United States Government developed the Federal Information Processing Standards (FIPS) to define procedures, architecture, algorithms, and other techniques used in computer systems. FIPS 140-2 is a government standard for encryption and cryptographic modules where each individual encryption component in the overall solution requires an independent certification.

Federal Information Processing Standard 140-2 specifies requirements for hardware and software products that implement cryptographic functionality. FIPS 140-2 is applicable to "all Federal agencies that use cryptographic-based security systems to protect sensitive [but unclassified] information in computer and telecommunication systems (including voice systems) as defined in Section 5131 of the Information Technology Management Reform Act of 1996, Public Law 104-106." The "-2" in FIPS 140-2 denotes the revision of the standard.

The full FIPS text is available online from the [National Institute of Standards and Technology](#) (NIST).

FIPS cryptographic modules and certification

FIPS compliance requires an explicitly defined continuous perimeter that establishes the physical bounds of a cryptographic module.

The *cryptographic boundary* defines this perimeter and contains the set of hardware, software, and firmware that implements valid security functions. Only the approved set of interfaces can access the cryptographic modules inside the cryptographic boundary. No other mechanism is allowed or provided when in FIPS mode.

Modules in the boundary perform these processes:

- FIPS-validated security methods performing cryptography, hashing, and related services running in ePO - On-prem
- Startup and verification testing required by FIPS
- Extension and executable signature verification
- TLS connection management
- Cryptographic API wrapping

i Important

Some older versions of Trellix products use non-FIPS-compliant ways to access ePO - On-prem cryptography and hashing services. Because these products violate the cryptographic boundary, they can't be used in FIPS mode. Check new versions of Trellix products for further information about FIPS compliance as they are released.

Trellix leverages these cryptographic modules to meet the requirements for FIPS-compliance.

Validated FIPS 140-2 cryptographic modules used by ePO - On-prem

Cryptographic module	Certificate number	Link
RSA BSAFE Crypto-C Micro Edition (Crypto-C ME) 4.1.2	2294	https://csrc.nist.gov/projects/cryptographic-module-validation-program/Certificate/2294
Bouncy Castle FIPS Java API (BC-FJA) 1.0.2.3	3514	https://csrc.nist.gov/projects/cryptographic-module-validation-program/certificate/3514
OpenSSL FIPS Object Module 3.1.2  Note: This module is used only for TLS communication between ePO - On-prem and the Trellix Agent.	2398	https://csrc.nist.gov/projects/cryptographic-module-validation-program/Certificate/2398

For ePO - On-prem 5.10.0 Service pack 1 Update 5, FIPS 140-3 modules are recommended.

Validated FIPS 140-3 cryptographic modules used by ePO - On-prem

Cryptographic module	Certificate number	Link
Bouncy Castle FIPS Java API (BC-FJA) 2.0.0	4743	https://csrc.nist.gov/projects/cryptographic-module-validation-program/certificate/

Cryptographic module	Certificate number	Link
		4743

ePO - On-prem operating modes

Depending on your environment and installation choices, ePO - On-prem operates in FIPS mode or Mixed mode.

The mode that a ePO - On-prem server runs in is determined during installation or upgrade and can't be changed.

FIPS mode

A ePO - On-prem server runs in FIPS mode after a clean installation with FIPS mode enabled.

In FIPS mode, ePO - On-prem:

- Places extra constraints on the types of security methods allowed
- Performs additional tests on startup
- Allows connections only from a FIPS-compliant version of the Trellix Agent

Your organization might need to use ePO - On-prem in FIPS mode if you fall into one of these categories:

- You are a US Government organization required to operate FIPS 140-2 compliant cryptographic models per FISMA or other Federal, State, or local regulations.
- Your organization requires the use of standardized and independently evaluated cryptographic modules per Company policy.

Don't use ePO - On-prem in FIPS mode if you fall into one of these categories:

- You integrate with legacy systems or products that do not support ePO - On-prem in FIPS mode.
- Your organizational policies allow you to choose which products or cryptographic modules to operate in FIPS mode. For example, an organization might elect not to operate ePO - On-prem in FIPS mode, and only operate Trellix on mobile computers in FIPS mode.

Mixed mode

For detailed instructions on configuring or recovering these modes

This mode is a standard ePO - On-prem installation not running in FIPS mode.

In Mixed mode, ePO - On-prem does not follow the constraints and tests described for FIPS mode, and is not compliant with FIPS levels of security.

Note

Your managed systems are still secure, but the certificates and Secure Sockets Layer (SSL) and Transport Layer Security (TLS) protocols are different.

For specific steps on installation, upgrades, and disaster recovery in FIPS mode, see:

- **Installation:** [Install ePO - On-prem in FIPS mode](#)
- **Upgrading:** [Upgrade from an earlier FIPS-compliant ePO - On-prem server](#)
- **Disaster recovery:** [Restoring ePO - On-prem server in FIPS mode](#)

To confirm that your components are correctly enforcing FIPS 140-2 mode, see [FIPS 140-2 operational status in ePO](#).

Install ePO - On-prem in FIPS mode

FIPS mode installation requires that you run the `Setup.exe` installer from the command line, adding a command-line option.

1. In a command window, change directories to the folder that include the ePO - On-prem installer.
2. Invoke the installer with the command `setup.exe ENABLEFIPSMODE=1`.
3. Continue with the installation.

Important

Do not change the default setting for the agent-server secure communication (ASSC) port. Leave it set as enabled on port 443. In FIPS mode, the agents communicate with the ePO - On-prem server using this ASSC secure port.

Upgrade from an earlier FIPS-compliant ePO - On-prem server

FIPS mode upgrades require you to run the `Setup.exe` installer from the command line, adding a command-line option.

If your existing ePO - On-prem server isn't running in FIPS mode, perform a complete reinstallation to change to FIPS mode.

 Important

When you install ePO - On-prem in FIPS mode, you can't restore a ePO - On-prem database from a previous non-FIPS ePO - On-prem server.

1. In a command window, change directories to the folder with the new ePO - On-prem installer.
2. Invoke the installer with the command **setup.exe ENABLEFIPSMODE=1**.
3. Continue with the upgrade.

Restoring ePO - On-prem server in FIPS mode

You can restore a ePO - On-prem server in FIPS mode only if the server was previously running in FIPS mode.

You can't restore a ePO - On-prem server that wasn't in FIPS mode as a FIPS mode ePO - On-prem server. The ePO - On-prem software and database must be reinstalled as a new instance of ePO - On-prem.

The complete ePO - On-prem reinstallation is required because all existing signed and encrypted content was signed with non-FIPS mode keys. Also, the database contains content encrypted with non-FIPS mode keys and can't be decrypted with the FIPS mode keys.

Verify FIPS 140-2 operational status in ePO

To maintain a compliant environment, you must validate that the core Trellix ePO - On-prem components are operating strictly within the cryptographic boundary.

You must verify the configuration of the following three components to confirm they are correctly enforcing FIPS 140-2 constraints and have disabled non-compliant "Mixed mode" operations:

Verify the Agent Handler status in FIPS 140-2 mode

View the server.ini file to make sure that Agent Handler is running in FIPS mode.

1. Use a text editor to open the server.ini file.

The server.ini file is located in your ePO - On-prem installation directory: <epoinstalldirectory>\DB\server.ini

2. Look for the FipsMode value.

This value indicates the server operating mode:

- FipsMode=0 — The server is in Mixed (normal) mode. To put your server in FIPS mode, repeat the installation or upgrade process.

- `FipsMode=1` — The server is in FIPS mode.

Verify the Apache Server configuration in FIPS 140-2 mode

The Apache server contains a FIPS enablement configuration setting.

1. Browse to the Agent Handler installation folder. The default folder is `C:\Program Files (x86)\Trellix\ ePolicy Orchestrator`.
2. Browse to the Apache configuration folder: `apache2\conf`
3. Using a text editor, open the `httpd.conf` file and search for `SSLFIPS`.
 - Off — Apache `mod_ssl` is not configured for FIPS enablement.
 - On — Apache `mod_ssl` is configured for FIPS enablement.

Verify the Application Server in FIPS 140-2 mode

View the Security Mode to make sure that the ePO - On-prem application server is running in FIPS mode.

Select Menu → Configuration → Server Settings → Security Keys, then confirm that Security Mode is FIPS 140-2.

Uninstall ePO - On-prem

Uninstalling the ePO - On-prem software requires specific consideration of your database.

If you intend to reinstall ePO - On-prem software later, and want to manage agents deployed by the current installation, back up your agent-server communication keys. You can't regenerate these keys later.

1. Close all database management software.
2. On the system where your ePO - On-prem server is installed, open the Windows Control Panel, then click Programs and Features → Trellix ePolicy Orchestrator-On-prem → Uninstall/Change.

The Remove Trellix ePolicy Orchestrator-On-prem dialog box opens.

3. Select whether to Also remove the ePolicy Orchestrator-On-prem database, then click Remove.



Supply credentials to grant sufficient permissions to remove the database. If the provided credentials are not sufficient, you can complete the uninstall process without removing the database.

Uninstall ePO - On-prem from a cluster

Uninstalling ePO - On-prem from a cluster environment requires that you take specific steps, depending on which server-class operating system you are running.

1. To set all ePO - On-prem services to offline, open the Windows Cluster Administrator/Management tool, then click Start → Programs → Administrative Tools → Failover Cluster Manager.
2. In the ePO - On-prem application group, right-click each of the ePO - On-prem resources and select Delete.
3. To uninstall the software, click Programs and Features → Trellix ePolicy Orchestrator-On-prem → Uninstall/Change.
4. Repeat this task on each node in your cluster.

Uninstall Agent Handler from ePO - On-prem

You can successfully uninstall Agent Handler from your system using the following methods.

- [Uninstall through Windows Control Panel](#)
- [Uninstall using the installer file](#)
- [Perform silent uninstallation](#)

Important

If you want to reinstall the Agent Handler later and manage existing deployed agents, ensure you back up your agent-server communication keys before proceeding, as they cannot be regenerated later.

Uninstall through Windows Control Panel

1. Open the Windows Control Panel on the system where your server is installed.
2. Navigate to Programs and Features.
3. Select Trellix Agent Handler → Uninstall/Change.

Uninstall using the installer file

1. Locate the Agent Handler Installer folder.
2. Run `SetUp.exe`.
3. Choose one of the following options:
 - Remove: Select this option to completely uninstall the Agent Handler.
 - Modify: Use this option to change the ePO Server details or reconfigure Remote Agent Handler to another ePO.
 - Update the necessary ePO Server details.
 - Enter Admin username and password and click Next.
 - Enter the database credentials.
 - Click Next to apply the modifications.
 - Repair: Select this option if you want to repair an existing installation.
 - Enter Admin username and password and click Next.
 - Enter the database credentials.
 - Click Next to initiate the repair process.

Silent uninstallation through command prompt

1. Open a command prompt in Administrator mode.

2. Run the following command:

```
Setup.exe /qbx EPO_Server={server_info} MFSADMINUSERNAME_UE={User_Name}  
MFSADMINPASSWORD_UE={Password} INSTALLDIR={AH Install Dir}  
SQL_SERVER={EPO_DB} SQL_DATABASE={EPO_DBName} SQL_USER=sa  
SQL_PASSWORD={Sql_Password}
```

COPYRIGHT

Copyright © 2026 Musarubra US LLC.

Trellix and FireEye are the trademarks or registered trademarks of Musarubra US LLC, FireEye Security Holdings US LLC, and their affiliates in the US and /or other countries. Other names and brands are the property of these companies or may be claimed as the property of others.

